

Structure-based drug design; Computational strategies in drug discovery; Antihypertensive agents; Antiviral drugs; Molecular docking; QSAR; Pharmacological insights

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ABSTRACT

This review meticulously examines the development, design, and pharmacological assessment of both well known antiviral and antihypertensive medications all time employing new chemical techniques and structure-based drug design to design and synthesize vital therapeutic entities such as aliskiren (renin inhibitor), captopril (a2-ACE-Inhibitor), dorzolamide (inhibitor of carbonic anhydrase) the review demonstrates initial steps regarding the significance of stereoselective synthesis, metal chelating pharmacophores, and rational molecular properties. More importantly, protease inhibitors (i.e., saquinavir, ritonavir, indinavir, amprenavir, etc.) and more contemporary agents (i.e., oseltamivir, nirmatrelvir/ritonavir (Paxlovid), etc.) were identified in the review and form a basis of advancement in antiviral therapy. Using the above-mentioned computational applications (i.e., molecular dockings, structure–activity relationships (SAR)), quantitative structure-activity relationships (QSAR modeling), and ADMET profile information, selectivity, safety, and therapeutic reliability of compounds toward antiviral activity and antihypertensive activity have improved further as outcomes of drug discovery research. Through the combinatorial application of computational drug design and experimental chemistry, researchers have significantly optimized the drug discovery process, minimized off-target effects, and expedited the pathway to develop therapeutically useful medications. In conclusion, this review highlights the transformative power of interdisciplinary approaches, including structure-based design, computational modeling, and the emerging approaches of drug repurposing and virtual screening. This hybrid approach provides a greater opportunity to improve drug therapy for cardiovascular disease and viral infections, and represents the rewards of collaboration.

1. Introduction

The ongoing emergence of complicated diseases and viral outbreaks creates a continuing need for new, improved, selective, and safer therapeutic agents. Over the last few decades, advancements in medicinal chemistry, computational biology, and molecular pharmacology have converged to create a new paradigm for drug discovery processes and, subsequently, drug development. Some flags of hope on the horizon of drug development are SBDD (Structure-Based Drug Design) and CADD (Computer-Aided Drug Design). These advances have revolutionized many aspects of drug discovery and development, and repositioned them to be accurate and efficient, and ultimately, the development pipelines of therapies are being so much cheaper (Niazi and Mariam, 2023; Poroikov, 2020a; Mohammed et al., 2025; Fahim, 2024). Notably, cardiovascular diseases, comprising hypertension, remain the leading

cause of morbidity and mortality worldwide. The advances in the clinical treatment of hypertension in the form of molecularly-targeted drugs, ACE inhibitors, carbonic anhydrase inhibitors, direct renin inhibitors, and the like have made significant contributions to the treatment of hypertension (De Clercq, 2002; Tolan et al., 2025; Dacrory and Fahim, 2020). Captopril, dorzolamide, and Aliskiren are all molecularly targeted medications; the structure of the enzymes and active sites of the enzymes have engendered rational drug optimization to afford newer classes of medication with increased safety, potency, selectivity, pharmacokinetic profiles, and the like. At the same time, the global burden of viral illnesses that have and will continue to increase, such as HIV, influenza, and hepatitis infections, most recently includes the SARS-CoV-2 pandemic. There is a growing need for antiviral medicines (Sun et al., 2022; Elsayed and Fahim, 2025; Shalaby et al., 2024; Fahim et al., 2019). Categorical viral drug families, such as protease inhibitors

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(saquinavir, ritonavir, indinavir, amprenavir, and even in development). and neuraminidase inhibitors (oseltamivir, and the more recent protease-targeting combinations such as nirmatrelvir/ritonavir (Paxlovid) as examples of successful drugs using structural biology and in silico modeling to develop effective antiviral agents to impact viral replication and improve clinical outcomes (Mahmoud and Fahim, 2025; Kheder et al., 2025; Khodair et al., 2025; Fahim et al., 2025; Fahim and Abas, 2025; Aboelnaga et al., 2023; Bayrak et al., 2023; Fahim et al., 2024; Fahim and Farag, 2020; Hashem et al., 2023; Fahim and Magd, 2021).

This review aims to bring together synthetic methodologies, structure-based design, and biological profiles of select antihypertensive and antiviral agents. This review highlights the integration of synthetic organic chemistry and modeling tools, including molecular docking, quantitative structure-activity relationships (QSAR), and ADMET determinants, which have simultaneously accelerated the transition from lead development to clinical practice. In addition, we discuss the increasing role of computational methods that address drug repurposing and virtual screening, which have been indispensable for quickly responding to new pathogens and finding new indications for existing drugs.

2. Huge Data for the development of novel drugs

The development of novel drugs is guided by several fundamental principles. These include understanding diseases caused by disruptions in biological processes, analyzing the regulatory signaling pathways that control normal and pathological cellular functions, and identifying molecular targets that respond to drug action to restore physiological balance. Additionally, drug discovery involves designing pharmacological agents that interact with these targets to achieve the desired therapeutic effect. This process also encompasses the optimization of drug-like molecules, including the development of new pharmaceutical compounds, to enhance therapeutic efficacy and safety (Niazi and Mariam, 2024; Poroikov, 2020b). Additionally, clinical medications that have been evaluated and approved for human use by regulatory authorities, such as the FDA, EMA, or other national agencies, are included in the treatment, prevention, or management of specific diseases and disorders (Fig. 1), and there are some of approved antihypertensive and antiviral drug as displayed in Table1.

Table1

Graphical timeline for the discovery and approval of selected antihypertensive and antiviral drug candidates from 1925–2021.

Year	Drug approval
1925	Captopril (ACE inhibitor) approved
1995	1995 – Dorzolamide (carbonic anhydrase inhibitor) approved
1996	1996 – Ritonavir, Indinavir, Saquinavir (HIV protease inhibitors) approved
1999	1999 – Oseltamivir (neuraminidase inhibitor) approved
2007	2007 – Aliskiren (direct renin inhibitor) approved
2018–2021	2018–2021 – Recent antiviral approvals and COVID–19 treatments, including Nirmatrelvir/ritonavir (Paxlovid)

3. Examples of drugs

3.1. Enzyme (ACE) inhibitors

3.1.1. Captopril (Capoten®, Bristol Myers-Squibb)

The angiotensin-converting enzyme (ACE) inhibitors are often prescribed medication for managing high blood pressure (hypertension), diabetic-related kidney disorders, and heart failure. As the first ACE inhibitor approved for use in clinical, it continues to play a vital role in cardiovascular therapy, offering significant benefits in controlling blood pressure and improving heart and kidney function (Ferguson and Vlasses, 1981).

3.1.2. Synthesis of captopril

The optically active compound (S)-3-hydroxy-2-methylpropanoic acid (1) undergoes a reaction with thionyl chloride (2) yielding the corresponding (S)-3-chloro-2-methylpropanoic acid (3). This intermediate readily couples with L-proline (4) to form (((S)-3-chloro-2-methylpropanoyl)oxy)-L-proline (5), which is subsequently converted into captopril (6)(yield=50 %) through a reaction with either hydrosulfide or trithiocarbonate ion in hot water, maintaining the stereochemistry throughout the process (Shimazaki et al., 1982). as shown in Scheme 1.

3.1.3. CADD of the development of Captopril and its SAR

✓ Rational Drug Design Based on ACE Structure

Captopril was developed through Structure-Based Drug Design (SBDD), taking advantage of knowledge of the structure and active site of angiotensin-converting enzyme (ACE). Researchers initially studied the natural ACE inhibitory peptides found in the venom of a species of snake called *Bothrops jararaca*. By analyzing the molecular

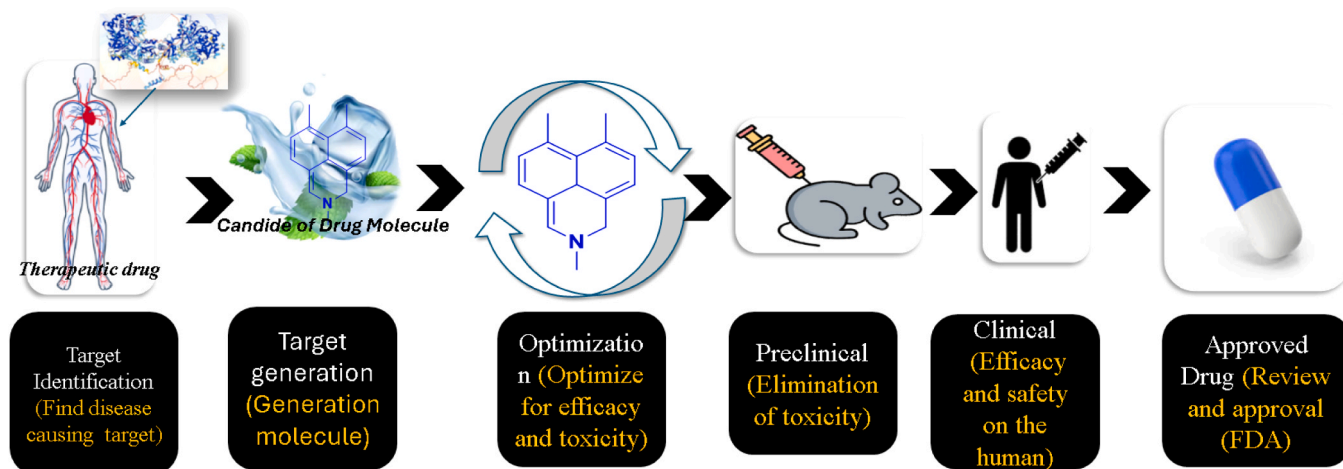
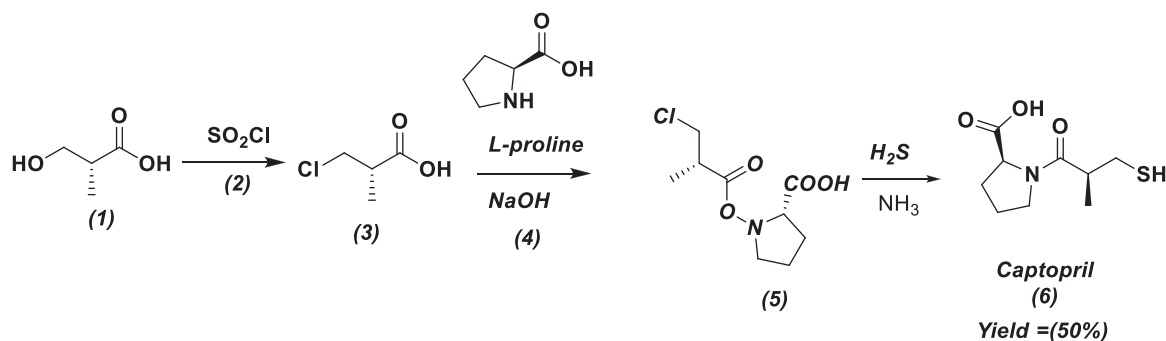


Fig. 1. The series of chemical and biological interactions that underpin modern drug discovery and development. The scheme outlines the process of understanding the disease, understanding the molecular target of interest, optimizing lead compounds, and conducting preclinical and clinical evaluation, resulting in approved antihypertensive and antiviral drugs.



Scheme 1. Synthesis of the Captopril drug.

interactions between these peptides and ACE, scientists identified key functional groups responsible for enzyme inhibition. This knowledge guided the rational design of captopril as a potent ACE inhibitor (Alberto-Silva et al., 2015; Camargo et al., 2012).

✓ Molecular Docking, Optimization, and QSAR:

The thiol (-SH) group in captopril was identified as a critical moiety for strong chelation with the zinc ion present in ACE's active site. This interaction significantly enhances inhibitory potency. Structural modifications were explored to maximize binding affinity while maintaining

pharmacokinetic properties suitable for clinical use. An isosteric methylene group takes the role of the amino functionality in benzyl succinic acid. Benzyl succinic acid was used as a model molecule to suggest that succinyl amino acids could also act as ACE byproduct inhibitors (Cushman et al., 1977; Ondetti et al., 1977; Byers and Wolfenden, 1973). Also, to catalyze captopril by mimicking the naturally occurring methyl group on the L-Ala residue in Ala-Pro, a stereospecific R-methyl group was first added to the succinyl group, resulting in the replacement of the carboxylic residue of succinic acid with a more potent zinc-coordinated mercapto activity (Ferreira et al., 1970) as displayed in Fig. 2(A) and

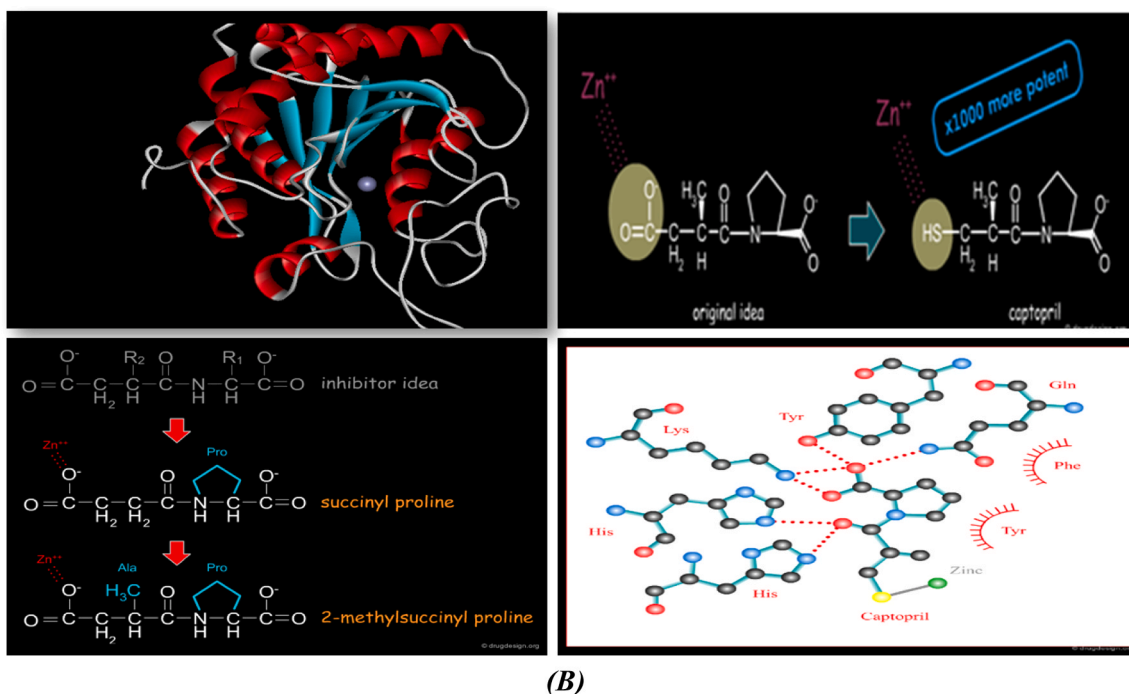
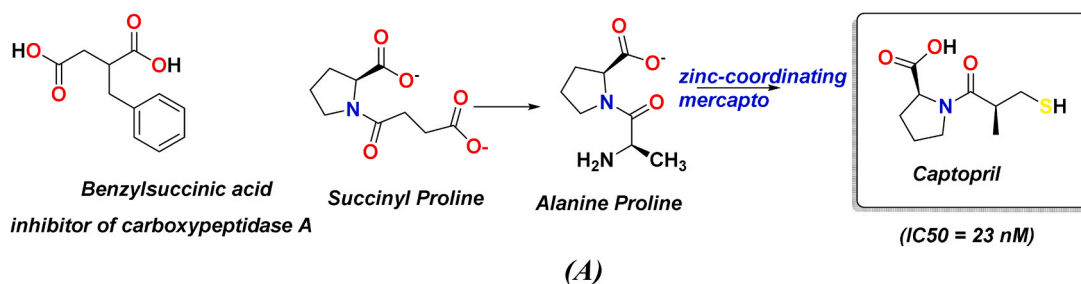


Fig. 2. (A) Rational design and synthetic approach to develop captopril, the first angiotensin-converting enzyme (ACE) inhibitor (B) Molecular modeling of ACE's active site and the orientation of the thiol group in captopril coordinating to the catalytic zinc ion, resulting in effective enzyme inhibition.

Captopril concurrently deactivates the vasodilator peptide bradykinin and prevents angiotensin I from being converted to the powerful vasoconstrictor angiotensin II. Captopril, developed by Cushman and Ondetti, was the first ACE inhibitor to enter use in clinical after receiving FDA approval in 1981. In the process of optimizing this lead, Cushman and Ondetti synthesized over 60 distinct analogs and evaluated their ability to block ACE. They eventually found the powerful ACE inhibitor, Captopril, which is depicted here. The enzyme was inhibited 1000 times more effectively by substituting a thiol group for the carboxylate, which was expected to interact with the zinc atom, than by 2-methyl succinyl proline, and 10–20 times more effectively by the venom peptide teprotide, as shown in Fig. 2(B) (Kitchen et al., 2004). Also, the (QSAR) modeling was used to predict and evaluate the potency of potential ACE inhibitors before synthesis to reduce the need for extensive experimental screening (Patchett et al., 1980).

In addition to the significant chelation of Zn^{2+} ions, MD (molecular dynamics) simulations highlight the gainful binding dynamic of Captopril to ACE. Captopril's methyl and proline groups are able to make small conformational shifts to maximize the van der Waals contacts within the S1' and S2' subsites of ACE, both hydrophobic. Thus, in MD simulations Captopril binding can displace an important water molecule from ACE's catalytic mechanism and essentially freezes the enzyme into its inactive conformation. This dynamic state illustrates how the static structure data ultimately leads to potent inhibition through both direct metal chelation and induced-fit inhibition of the enzyme's catalytic hydrolytic function.

This computationally-directed design was corroborated through impressive clinical outcomes. Captopril demonstrated marked reductions in blood pressure and mortality in patients with heart failure in large-scale clinical trials, ultimately becoming a first-line treatment and supporting the predictive validity of the initial SBDD method (Ferguson and Vlasses, 1981)

3.1.4. Side effect

Captopril exhibits a relatively unfavorable pharmacokinetic profile, characterized by a short half-life of 2–3 h and a duration of action ranging from 12 to 24 h. Due to its rapid elimination, multiple daily doses (two to three times per day) are required, which may affect patient adherence to the prescribed treatment regimen. Additionally, when administered during the second and third trimesters of pregnancy, captopril poses a risk of fetal harm, including serious injury and potentially fatal outcomes (Heel et al., 1980).

3.2. Carbonic anhydrase inhibitor

3.2.1. Dorzolamide (Trusopt®, Merck)

Dorzolamide, a potent carbonic anhydrase inhibitor (CAI), is widely prescribed for reducing intraocular pressure (IOP) in patients with glaucoma and ocular hypertension. Formulated as a topical ophthalmic solution, it directly inhibits aqueous humor production within the eye, delivering therapeutic benefits while minimizing systemic side effects (Balfour and Wilde, 1997; Kobayashi and Naito, 2000). Dorzolamide 2 % ophthalmic solution, the first topical carbonic anhydrase inhibitor, is widely used for the treatment of ocular hypertension and glaucoma by effectively lowering intraocular pressure (IOP). When administered three times daily, it reduces IOP by 4–6 mmHg at peak and 3–4.5 mmHg at trough. Dorzolamide has been shown to be as effective as pilocarpine 2 % (four times daily) as an adjunct therapy in patients receiving timolol and demonstrates additive ocular hypotensive effects when combined with topical beta-adrenergic antagonists. Notably, unlike some other IOP-lowering agents, dorzolamide is not contraindicated in patients with heart disease, reactive airway disease, or asthma, making it a safer alternative for individuals with these conditions (Balfour and Wilde, 1997; Konstas et al., 2003).

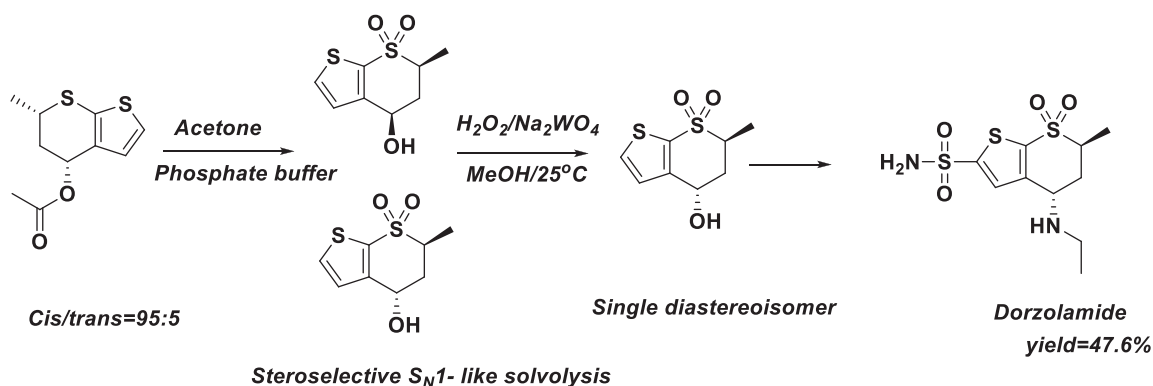
3.2.2. Synthesis of Dorzolamide

The synthesis of Dorzolamide proceeds through a stereoselective SN_1 -like solvolysis of the bicyclic thioketal precursor in acetone and a phosphate buffer, which leads to the characteristic (cis/trans $\approx$ 95:5) diastereomeric ratio, and provides high stereo control of that part of the synthesis and resulting in the generation of the chiral site necessary for selective biological activity. The remaining alcohol is mildly oxidized by hydrogen peroxide in the presence of sodium tungstate (Na_2WO_4) in methanol at ambient temperature to produce the sulfone. Not only does this pyrolidinium stabilization of the heterobicyclic framework activate the sulfur center for nucleophilic substitution later on, but the oxidation reaction rightly leads to the introduction of the sulfonamide pharmacophore through nucleophilic amination with ammonia (or suitable amine source). Overall, this route is an efficient scheme to combine stereoselective control and wet oxidation and amination steps to assemble Dorzolamide in good overall yield (47.6 %) - ready for several biological applications - all three steps were important to maintain structural and functional fitness of the drug molecule whilst attaining good yield in a short period as demonstrated in (Scheme 2) (Martinelli et al., 2023).

3.2.3. CADD of the development of Dorzolamide and its SAR

Dorzolamide was developed using SBDD by targeting human carbonic anhydrase II (CA-II), a metalloenzyme responsible for the production of bicarbonate in the eye, which regulates aqueous humor secretion and intraocular pressure (IOP). The above-mentioned iterative approach has enabled the optimization of carbonic anhydrase inhibitors to maximize potency in a structural series with a proper balance of aqueous solubility and lipophilicity, allowing topical use in the treatment of an ocular disease called glaucoma. The initial X-ray crystallography analysis of carbonic anhydrase II (CA-II), specifically Human CA II (HCA II), revealed a cone-shaped active-site cavity that narrows significantly near the catalytic hydrophobic pocket. In this region, zinc is tetrahedrally coordinated with three histidine residues: His94, His119, and His96. The active-site cavity is amphiphilic, with hydrophilic residues predominating on one wall and hydrophobic residues on the other (Ashburn and Thor, 2004; Oprea and Overington, 2015; Beck et al., 2020; Richardson et al., 2020; Lionta et al., 2014) which reveals a zinc-containing active site to design molecules capable of coordinating with the Zn^{2+} ion, essential for enzyme inhibition then makes docking simulation between the model the interaction between potential inhibitors and CA-II and the Dorzolamide drug optimized to fit deeply into the active site, ensuring strong and selective inhibition due to presence of sulfonamide ($-SO_2NH_2$) group which is crucial pharmacophore for CA-II inhibition, directly interacting with the Zn^{2+} ion. Also, the addition of hydrogen bonding and hydrophobic interactions improved binding affinity and specificity (Baldwin et al., 1989). HCA II was used to prepare and co-crystallize all four potential optical isomers, and the structure of each complex was determined using X-ray crystallography (Greer et al., 1994; Supuran, 2008) as displayed in Fig. 3(A). The isomer with the highest inhibitor efficacy and affinity contained the trans-S, S structure and a K_i value of 0.37 nM. The primary conformational difference among the four optical isomers bound within the active site, according to structural data from X-ray crystallography, was discovered to be the thiophene sulfonamide N-S-C-S dihedral angle. The S, S-isomer's torsion angle was 140° , while the R, R-isomer's was 175° (Greer et al., 1994; Cohen, 2007). An X-ray crystallographic analysis revealed that the 6-methyl group was directly above a lipophilic groove, one of whose walls was formed by Phe131. X-ray crystallography using this iterative method revealed that the side chain of His64 rotated 3.1 Å from its position in the native enzyme. Baldwin et al (Baldwin et al., 1989). and Greer et al (Greer et al., 1994). demonstrated by additional SAR and structural evidence that a 4-alkylamino group was responsible for the rotation of the His64 side chain (Leelananda and Lindert, 2016) as demonstrated in Fig. 3(B).

The X-ray experiment highlights a major difference in the thiophene



Scheme 2. Preparation of the Dorzolamide drug.

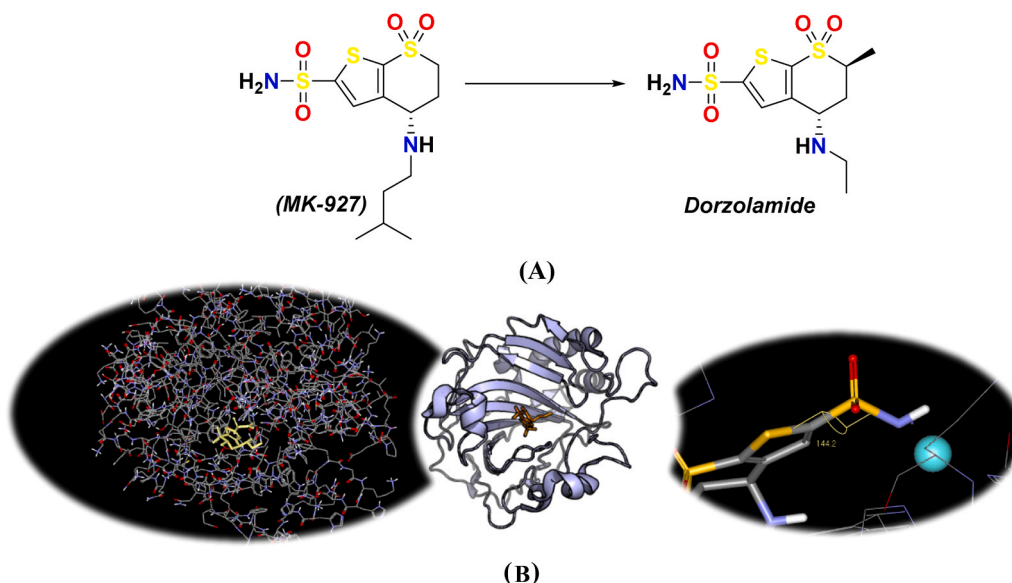


Fig. 3. (A) The comparative chemical structures of MK-927 and dorzolamide, highlighting key pharmacophoric features. (B) X-ray crystallographic structure of dorzolamide binding in the catalytic pocket of human carbonic anhydrase II (HCA II, PDB ID: 4M2U) with coordination of sulfonamide- Zn^{2+} and hydrophobic interactions with active site residues.

sulfonamide dihedral angle between active and inactive isomers. The MD simulations of dorzolamide bound to CA-II indicate that this optimal angle of 140° is stable, resulting in the sulfonamide group being properly positioned for bidentate coordination with the Zn^{2+} ion. This angle also allows the thienothiopyran ring to form hydrophobic interactions with stable interactions with residues such as Phe131. The dynamics of the R, R-isomer, in contrast, demonstrate that the dihedral angle slides away from an optimal angle, suggesting poor coordination to the Zn^{2+} and lower binding affinity. These dynamic analyses also offer a mechanistic rationale for the marked difference in potency between the two stereoisomers.

3.2.4. Side effect

Dorzolamide may cause ocular discomfort, including stinging, burning, itching, and a bitter taste. It can also lead to anterior chamber shallowing, resulting in transient myopia. As a second-generation carbonic anhydrase inhibitor, dorzolamide minimizes the systemic side effects commonly associated with first-generation inhibitors such as acetazolamide, methazolamide, and dichlorphenamide (Delaney et al., 2002).

3.3. Direct renin inhibitors

3.3.1. Aliskiren(Novartis and Speedel)

Aliskiren is a direct renin inhibitor (DRI) and the first drug in its class, primarily prescribed for the treatment of hypertension (high blood pressure). It functions by directly inhibiting renin, a crucial enzyme in the renin-angiotensin-aldosterone system (RAAS), which plays a fundamental role in blood pressure regulation (Sharma et al., 2019; Lubel et al., 2008; Szczepanska-Sadowska et al., 2018).

3.3.2. Synthesis of Aliskiren

The reduction of 2-((4-methoxy-3-((3-propoxy)methoxy)phenyl)methyl)-3-methylbut-2-enoic acid (1) to its corresponding alcohol was followed by chlorination using thionyl chloride ($SOCl_2$) in pyridine, yielding the alkyl chloride derivative. This derivative served as a precursor for the Grignard reagent, which subsequently reacted with 5-oxotetrahydrofuran-2-carbonyl chloride (2) to efficiently produce the keto-lactone in excellent yield. The resulting keto-enol intermediate was selectively reduced with sodium borohydride ($NaBH_4$), generating a (3:1) diastereomeric mixture (4). Subsequent methylation of the hydroxyl groups, followed by displacement with sodium azide in the presence of catalytic 15-crown-5, led to the formation of the azido-

lactone with complete inversion of configuration. The azido-lactone then underwent ring opening in the presence of amine 21; however, due to its weak nucleophilicity, stoichiometric 2-hydroxypyridine was required to facilitate the reaction. Finally, reduction of the azide group yielded Aliskiren with yield (60 %) (Rüeger et al., 2000), as shown in Scheme 3.

3.3.3. CADD of development of Aliskiren and its SAR

The development of Aliskiren began with a focus on renin, a protease that converts angiotensinogen into angiotensin I, which is then converted into angiotensin II, a powerful vasoconstrictor. By inhibiting renin, Aliskiren effectively lowers angiotensin II levels, leading to vasodilation and a subsequent reduction in blood pressure. As part of an optimization process, over 1400 analogs from all four series were synthesized. Among them, a phenoxy analog, later identified as Aliskiren, exhibited superior physicochemical and pharmacokinetic properties, making it a candidate for further research. The hydrophobic long chain of Aliskiren enabled effective interaction with the target protein, as illustrated in Fig. 4. First, X-ray crystallography was employed to determine the three-dimensional structure of the target protein renin, providing crucial insights into its active site architecture. Subsequently, molecular docking simulations were conducted to evaluate the binding interactions between renin and potential small-molecule inhibitors, facilitating the identification of promising drug candidates. These simulations allowed for the visualization of how various functional groups could interact with key active site residues, such as Asp 74, Ser 34, and His 111, in renin. This detailed analysis provided the foundation for the design of Aliskiren. Aliskiren was designed to mimic the renin cleavage reaction's transition state by incorporating a hydroxymethyl group that interacts with the enzyme's Asp 74 residue. This approach not only improves the selectivity and potency of drugs but also reduces off-target effects (Vergara-Rodriguez et al., 2009). And its SAR was developed through Hydroxyethylene Isostere: This core takes the place of the scissile amide bond (Leu-Val) in angiotensinogen. The (S)-hydroxyl group takes place at the transition state and makes vital hydrogen bonds with the catalytic aspartates (Asp32/Asp215) of renin. Alkoxy Motif (P2/P3 Pocket): The methoxypropoxy chain was optimal to fill the large P3 pocket (hydrophobic) in renin, however, shorter and less lipophilic chains (methoxy, ethoxy, etc.) lead to dramatically decreased potency. Benzyl Group (P1 Pocket): The S-benzyl group fills the deep S1

specificity pocket, which typically holds the substrate's leucine side chain. Amide Core: The amide in the middle provides rigidity and correct spacing to allow the P2 and P1' substituents to have optimal spacing and for the N-isopropyl group in P2 and the tert-butyl in P1' to have favorable hydrophobic access to their respective sub-pockets.

Although the initial docking indicates interaction with Asp74, Ser34, and His111, it is important to carry out longer MD simulations to fully understand the high binding affinity of Aliskiren. These longer simulations demonstrate that the long and flexible hydrophobic chain of Aliskiren confers a form of conformational selection as it 'snakes' into an extended hydrophobic cleft of renin, which is not completely occupied by the natural substrate, angiotensinogen. This induced fit creates a large buried surface area, which is also a large contributor to the slow rate of dissociation and long residence time for Aliskiren; both of which are essential for its efficacy in vivo.

3.3.4. Side effects

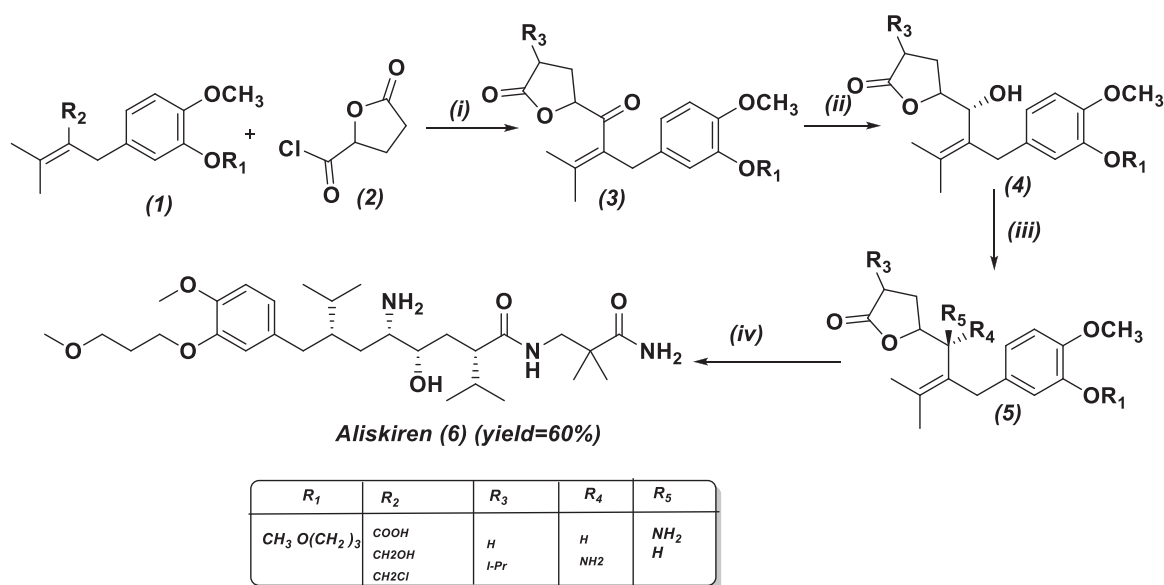
In Pregnancy: Aliskiren is contraindicated during pregnancy, as it can disrupt normal fetal kidney development. While Breastfeeding: is not recommended due to potential risks to the nursing, also in Cardiovascular Risk, it causes pre-existing kidney or heart disease (Parving et al., 2012).

3.4. Antiviral drugs

Antiviral drugs are specialist pharmaceuticals that battle viral infections by focusing on certain stages of the viral life cycle. Unlike antibiotics, which are effective against bacterial infections, antiviral agents work by either inhibiting viral replication or blocking the virus from entering and infecting host cells. These drugs serve an important role in the treatment and management of several viral infections, including HIV/AIDS, influenza, hepatitis, and herpes simplex, helping to control infection, reduce symptoms, and improve patient outcomes (De Clercq, 2002; Shiraki et al., 2012; De Clercq, 2004) (Kausar et al., 2021; De Clercq and Li, 2016; Levinson et al., 2020).

3.4.1. HIV drugs

3.4.1.1. Protease inhibitors. Antiretroviral medications are essential in the treatment of HIV, working to suppress viral replication and prevent



Scheme 3. Synthesis of Aliskiren drug (i) 13, Mg, BrCH₂CH₂Br, THF, rt; (ii) NaBH₄ (iii) MsCl, NEt₃, CH₂Cl₂, 82 ± 99 %, NaN₃, 15-crown-5, DMPU, 85 %; (iv) H₂ NCH₂C(Me)₂ CONH₂ (21), 2-OH-pyridine, NEt₃, 59 %; H₂, Pd: C MeOH, 74 %; NH₂ OHHCl, NaHCO₃, MeOH, rt, 81 %; NH₂ OBnHCl, NaHCO₃, 89 %; H₂, Pd: C, 1 N HCl, MeOH.

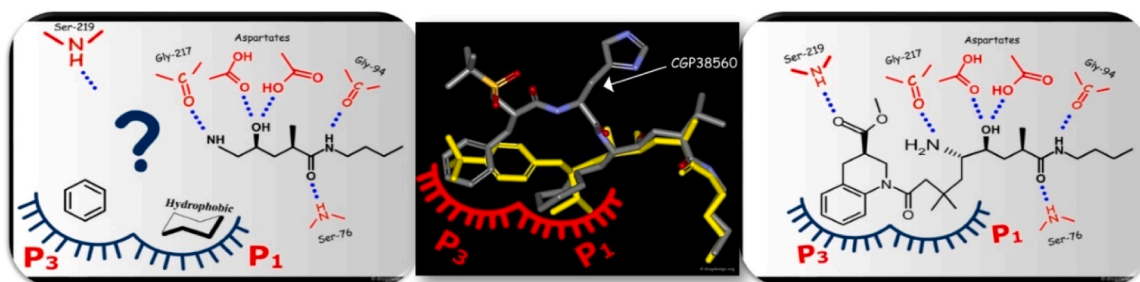


Fig. 4. Molecular interaction map of aliskiren with the active site of renin. The hydrophobic chains and aromatic groups of the ligand form extensive van der Waals and hydrophobic contact with renin. The hydroxymethyl group forms specific interactions with Asp74, Ser34, and His111, stabilizing binding and prolonging the presence time in the active site.

further damage to the immune system. By reducing the number of viruses in the body, these medications support immune recovery and slow disease progression. Given HIV's ability to rapidly mutate and develop resistance, effective treatment relies on a combination of different antiretroviral drugs (see Fig. 5). Among these, protease inhibitors (PIs) play a crucial role by targeting the HIV protease, that is required for the virus to produce mature, infectious particles. By inhibiting this enzyme, PIs disrupt the viral replication process, leading to a reduction in viral load and improved disease management (Anderson, 2003) (Leelananda and Lindert, 2016; Eckhardt and Gulick, 2017; Mandala et al., 2016; Brik and Wong, 2003). And the binding as displayed in Fig. 5 showed the dimeric HIV-1 protease (HIV PR) enzyme. Its role is to cleave the viral Gag and Gag-Pol polyproteins into the mature, functional proteins that are required for assembling new infectious viral particles. If this enzyme is inhibited, viral replication is stopped and the blue spheres show the hydrophobic extended active site cleft of the enzyme, which is located at the dimer interface. This cleft is symmetrically positioned and can accommodate extended polypeptide substrates with lengths of approximately 8 amino acids long. The central region of the cleft houses the catalytic aspartic acid residues (Asp25 from each monomer). As general Protease inhibitors (PIs), such as saquinavir, ritonavir, and indinavir, are peptidomimetics. They were developed to replicate the transitioning state of the natural peptide substrate during hydrolysis. They may have a transition state with a central non-cleavable core, rather than a scissile peptide bond (e.g., saquinavir/indinavir use a hydroxyethylene group), which is designed to allow tight binding via the catalytic aspartates. Their side chains (often large hydrophobic groups like phenylalanine and isobutyronitrile mimics) extend into parts of the substrate-enzyme subsites (S1, S2, S1', S2', etc.) and form significant van der Waals contacts with the enzyme: they fill hydrophobic pockets. These types of

high-affinity binding will competitively block the natural substrate from entering the site.

The inherent flexibility of binding poses important challenges to HIV protease inhibitor binding and susceptibility to resistance. Even before resistance occurs, MD simulations have helped illustrate how potential common mutations (e.g., V82A, I84V, L90M) affect the dynamics the binding pocket. Mutations often do not fully block or inhibit binding but rather create impairment of activity by changing the conformational landscape for binding, such as affecting flexibility of loops, preference for sidechain rotamer, and the water-mediated hydrogen-bonding network. This allows the viral protease to sample conformations and accumulate conformational states with reduced affinity for the drug compared to the natural substrate to establish clinical drug resistance. The goal for next-generation PI design is to fully understand what has dynamically transpired.

The use of PIs like saquinavir and ritonavir, with this structural knowledge, represented a milestone in the AIDS pandemic. Clinical trials showed that combination antiretroviral therapy with PIs could reduce viral loads to undetectable levels and provide life-saving virologic control that could reduce AIDS-related mortality, and proved that the structure-based approach was a valid one (Bahadur Gurung et al., 2020; Feld et al., 2014)

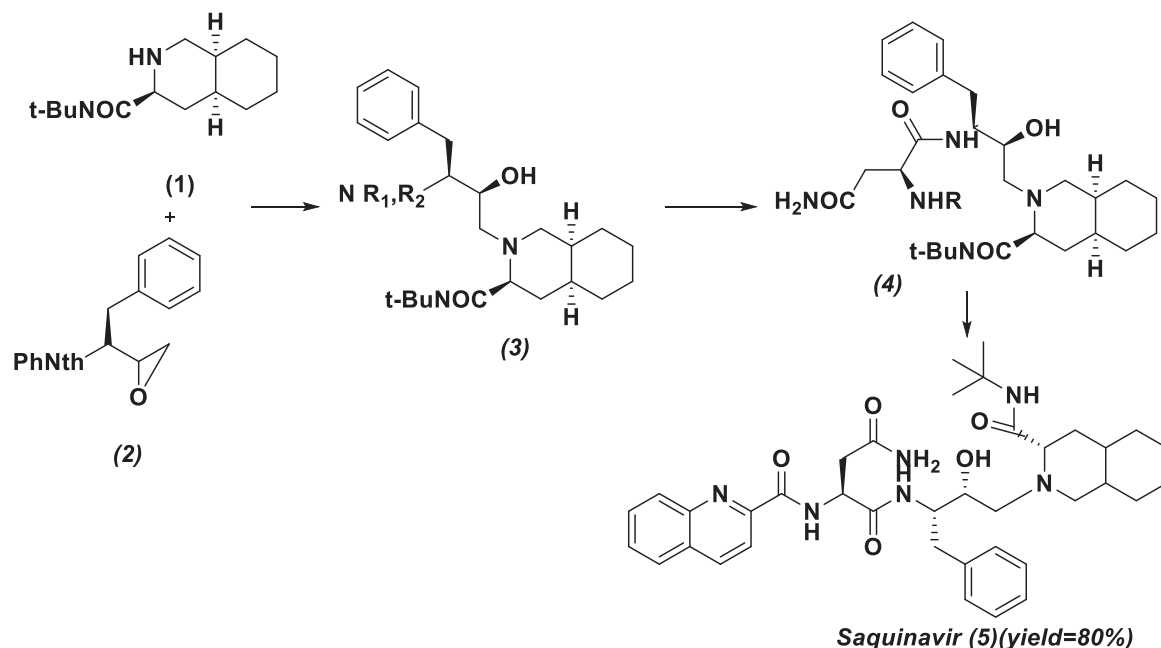
From these drugs as below:

3.4.1.2. Saquinavir(Roche). Saquinavir is an antiretroviral protease inhibitor that binds to the HIV protease enzyme's active site. Saquinavir inhibits this enzyme, which prevents the cleavage of viral precursors from polyprotein and thus the formation of mature, functional proteins required for viral replication. This disruption effectively impairs the production of infectious HIV particles, helping to lower viral load and controlling the advancement of the infection (Abad Martínez et al., 2014).

3.4.1.3. Synthesis of Saquinavir drug. Saquinavir was assembled via stepwise construction of a hydroxyethyl-urea transition-state mimic, followed by peptide-like couplings and deprotections. Initial condensation of the *t*-Boc-protected piperazine fragment of quinaldic acid derivative (1) with the phthaloyl-protected aminomethyl epoxide (2) followed by deprotection of the phthaloyl group to form hydroxyethyl-urea core (3') through nucleophilic addition to an activated carbonyl and urea formation; this key step typically proceeds in 77 % yield and establishes the secondary alcohol adjacent to the urea that mimics the tetrahedral intermediate of HIV protease catalysis. Selective carbamate (t-Boc) deprotection unveils the amine for chain extension (84 % yield), after which amide/urea coupling installs the bulky hydrophobic side chains to give protected Saquinavir (6') in 55–75 % yield. Final global/stepwise deprotections furnish Saquinavir (5) in 80 % yield, delivering material with the required stereochemical integrity for high-affinity binding in the protease active site (Wolfgang Göhring and Hilpert, 1996). as demonstrated in Scheme 4.



Fig. 5. Using SiteHound analysis to locate binding sites of HIV-1 protease. The blue spheres denote the primary inhibitor-binding pocket used by protease inhibitors (PIs), and serves as the structural basis for the design of antiretroviral agents.



Scheme 4. Steps for the synthesis of the Saquinavir drug in multistep reactions.

3.4.1.4. CADD of development of Saquinavir. The FDA approval of Saquinavir, an HIV protease inhibitor, was significantly advanced through structural modeling of the HIV protease using X-ray crystallography as PDBID: (3OXC) as displayed in Fig. 6, which revealed its active site and dimeric nature. This structural insight enabled the design of peptidomimetic inhibitors that target HIV polyprotein cleavage sites. To refine Saquinavir's efficacy, structure-activity relationship (SAR) studies, energy minimization, and molecular dynamics simulations were conducted to evaluate its stability and interaction within the enzyme's active pocket. Further optimization of the drug was achieved using Rule of Five of Lipinski, along with in silico ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) modeling. These modifications enhanced oral bioavailability and improved resistance to metabolic degradation. Saquinavir inhibits HIV protease by forming hydrogen bonds and hydrophobic interactions in the enzyme's active site. By mimicking the natural substrate, it effectively prevents the protease from cleaving viral polyproteins into functional components, thereby inhibiting viral replication (Kim et al., 1995). As displayed in the interaction in Fig. 6 Saquinavir bound to HIV-1 protease (pdb: 3OXC). Specific interactions: Saquinavir is a potent transition-state mimetic inhibitor. Its central hydroxyl group interacts with the catalytic aspartate dyad (Asp 25/Asp 25') of the HIV protease. This

interaction mimics the tetrahedral intermediate of the proteolytic reaction through hydrogen bonding with C=O and amide moieties in its backbone forming key hydrogen bonds with the backbone of the flap regions of the protease (Ile50/Ile50') and is a conserved feature that stabilizes the closed conformation of the enzyme upon inhibitor binding, and hydrophobic interactions: Its two decahydroisoquinoline groups and the quinoline moiety are intended to occupy the S1/S1' and S2/S2' hydrophobic subsites of the protease, respectively. This large number of van der Waals' contacts greatly increases its binding energy and is a point of specificity. So generally, this binding does a great job of effectively "plugging" the active site to prevent processing of viral polyproteins such as GagPol precursor proteins and allow only the production of immature, non-infectious virions.

3.4.1.5. SAR of Saquinavir drug. The development of saquinavir was elucidated through SAR studies that optimized its peptide-like structure to maximize binding affinity and metabolic stability. Central Hydroxyethylamine Core: This moiety is an important non-cleavable transition-state isostere that replaces the scissile amide bond in the natural substrate. It is also required for high-affinity binding to the catalytic aspartate dyad (Asp25/Asp25'). Modifications (including removal of the hydroxyl group) led to loss of inhibitory activity. Decahydroisoquinoline

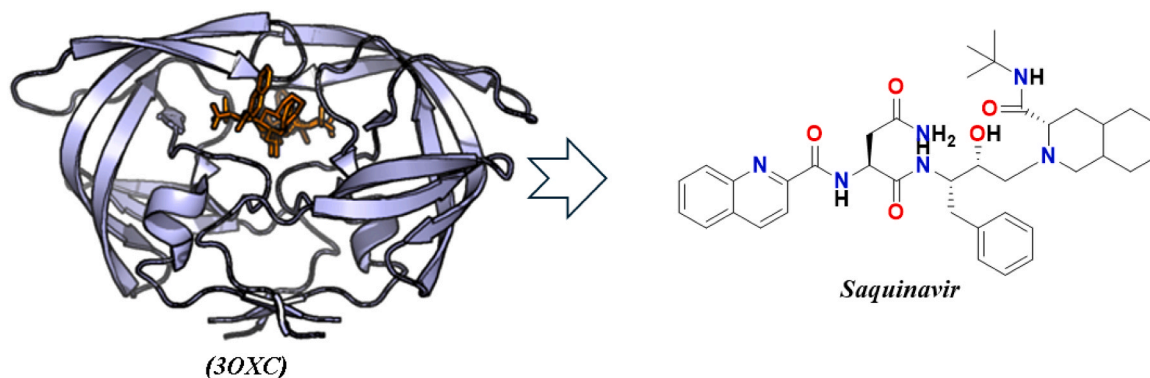


Fig. 6. In the molecular docking of saquinavir in dimeric HIV-1 protease (PDB ID: 3OXC), the binding mode shows H-bonds and hydrophobic contacts that mimic viral polyprotein substrates, blocking proteolytic cleavage and viral maturation.

(S1/S1' Subsites): The stereochemistry (S) of this fused bicyclic structure is essential for optimal fit into the hydrophobic S1 and S1' pockets of the protease. Enlarging or reducing the ring systems or altering the stereochemistry led to a significant loss in potency. Quinoline-2-carbonyl Group (S2 Subsite): This large, hydrophobic moiety can fill the large S2 subsite. SAR confirmed that this moiety's planarity and lipophilicity contributed most to the binding affinities. Less lipophilic or smaller groups reduced activity. tert-Butylamide (S2' Subsite): The tert-butyl was optimum for filling the shallow, hydrophobic S2' subsite. There was a loss of potency using smaller alkyl amides (e.g.: methyl, ethyl), reflecting the need for a specific steric bulk in this region.

3.4.1.6. Side effect. Saquinavir, a protease inhibitor used in the treatment of HIV infection, is associated with several adverse effects, including nausea, diarrhea, abdominal pain, fatigue, headache, and vomiting. In cases of excessive intake or prolonged use, it may lead to more serious complications such as liver toxicity, cardiovascular complications, metabolic disturbances, lipodystrophy syndrome, pancreatitis, and hypersensitivity reactions. Careful monitoring and appropriate dosage adjustments are essential to minimize these risks (von Hentig et al., 2007).

3.4.2. Ritonavir (Abbott company)

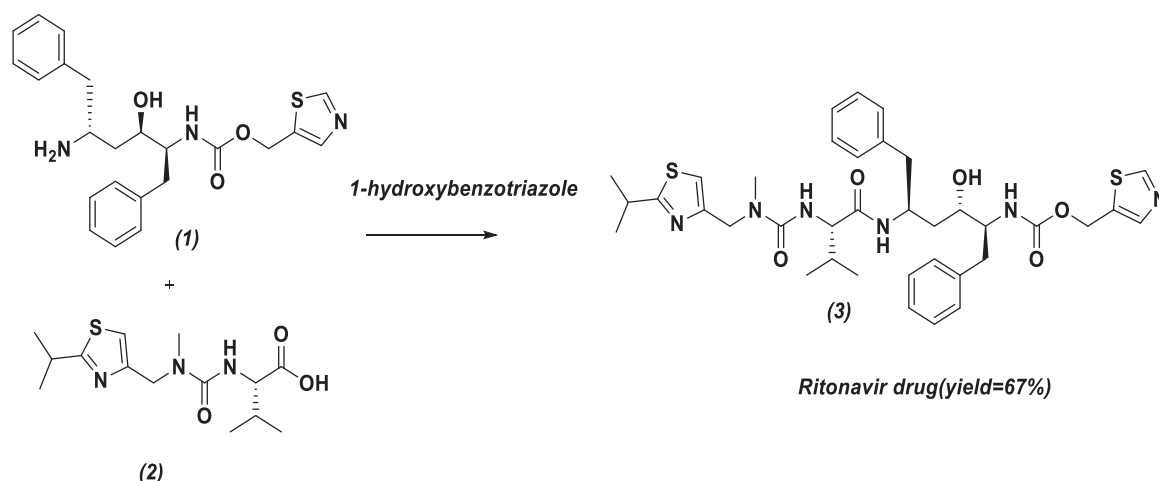
Ritonavir, a protease inhibitor (PI), plays a crucial role in the treatment of HIV/AIDS. Initially developed as an active antiretroviral agent, it is now primarily used as a pharmacokinetic enhancer, boosting the efficacy of other protease inhibitors by inhibiting their metabolism and prolonging their therapeutic effects (Findlay, 2007). Ritonavir functions as a potent inhibitor of the CYP3A4 enzyme, reducing the metabolism of co-administered antiretroviral drugs. This inhibition leads to higher plasma concentrations, improved bioavailability, and prolonged therapeutic effects. Mechanistically, Ritonavir binds to the active site of the HIV protease enzyme, preventing the cleavage of viral polyproteins into mature, functional proteins essential for viral replication. At low doses, its primary role is as a CYP3A4 inhibitor, significantly enhancing the potency and stability of companion antiretroviral therapies (Popović-Djordjević et al., 2022; Loos et al., 2023).

3.4.2.1. Synthesis of Ritonavir drug. The coupling reaction between thiazol-5-ylmethyl ((2S,3R,5R)-5-amino-3-hydroxy-1,6-diphenylhexan-2-yl)carbamate (1) and (((2-isopropylthiazol-4-yl)methyl) (methyl)carbamoyl)-L-valine (2) was carried out in the presence of 1-hydroxybenzotriazole hydrate (HOBt), 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide (EDC), and dicyclohexyl carbodiimide (DCC). The reaction was performed in tetrahydrofuran (THF) as a solvent, yielding the corresponding 1,3-

thiazol-5-ylmethyl N-[(2S,3S,5S)-3-hydroxy-5-[(2S)-3-methyl-2-[(methyl)({[2-(propane-2-yl)-1,3-thiazol-4-yl]methyl})carbamoyl]amino]butan-2-amido]-1,6-diphenylhexan-2-yl]carbamate (3) (Ritonavir drug) (yield=67 %) (Kempf et al., 1998) as displayed in Scheme 5.

3.4.2.2. CADD of production of Ritonavir and its SAR. The development of Ritonavir highlights the transformative role of computational techniques in modern drug discovery. Ritonavir functions as a competitive inhibitor of HIV protease (PDB ID: XK263), blocking the enzyme's ability to cleave viral polyproteins necessary for HIV replication. The dissociation of Ritonavir from HIV protease provides critical insights into its molecular interactions, shedding light on the energetic contributions governing its binding stability. The analysis of Ritonavir's unbinding process quantifies the interaction energies between the drug and the enzyme, where non-polar interaction energy is represented in red, polar interaction energy in black, and total interaction energy in blue. These interactions are evaluated. Root Mean Square Deviation (RMSD) distance offers a detailed view of the binding interface's dynamics and stability during dissociation. By integrating molecular modeling, structure-activity relationship (SAR) analysis, ADMET predictions, and docking simulations (as illustrated in Fig. 7), this study provides a comprehensive understanding of Ritonavir's binding affinity and unbinding kinetics, reinforcing its role in HIV protease inhibition (Sun et al., 2022). , also able to optimize its binding affinity, metabolic stability, and pharmacokinetic properties. Its dual functionality as both an HIV protease inhibitor and a CYP3A4 booster has made Ritonavir a cornerstone of antiretroviral therapy (ART) and an important treatment in managing COVID-19 (Kempf et al., 1998). Moreover, the interaction of Ritonavir Binding and Dissociation with HIV-1 Protease in Fig.7 showed the dual function of ritonavir as both a direct inhibitor of HIV protease and as a pharmacokinetic enhancer (CYP3A4 inhibitor). Firstly, binding HIV Protease: Ritonavir, like saquinavir, binds at the active site of HIV protease in a similar way as a competitive inhibition. Its structure has a central hydroxyl group, anchoring it to the catalytic aspartates, with hydrophobic side chains occupying the hydrophobic subsites on the enzyme. The analysis of its unbinding pathway (as derived from MD simulation) calculates the energetic cost of unbinding these interactions. The non-polar (red) energy cross-section illustrates the degree to which non-polar/hydrophobic contacts contribute to the binding free energy cost (as it is the dominant factor), while the polar (black) energy cross-section accounts for hydrogen bonding and electrostatic contacts.

Then CYP3A4 Inhibition with a potent inhibitor of the liver cytochrome P450 3A4 (CYP3A4) enzyme; an inhibition that can form a tight, quasi-irreversible complex with the heme iron in CYP3A4 and thus greatly slow the metabolism of co-administered medication drugs that have this as a substrate (e.g., other PIs like lopinavir). This "boosting"



Scheme 5. Coupling of carbamate derivative with L-valine derivative to give Ritonavir drug.

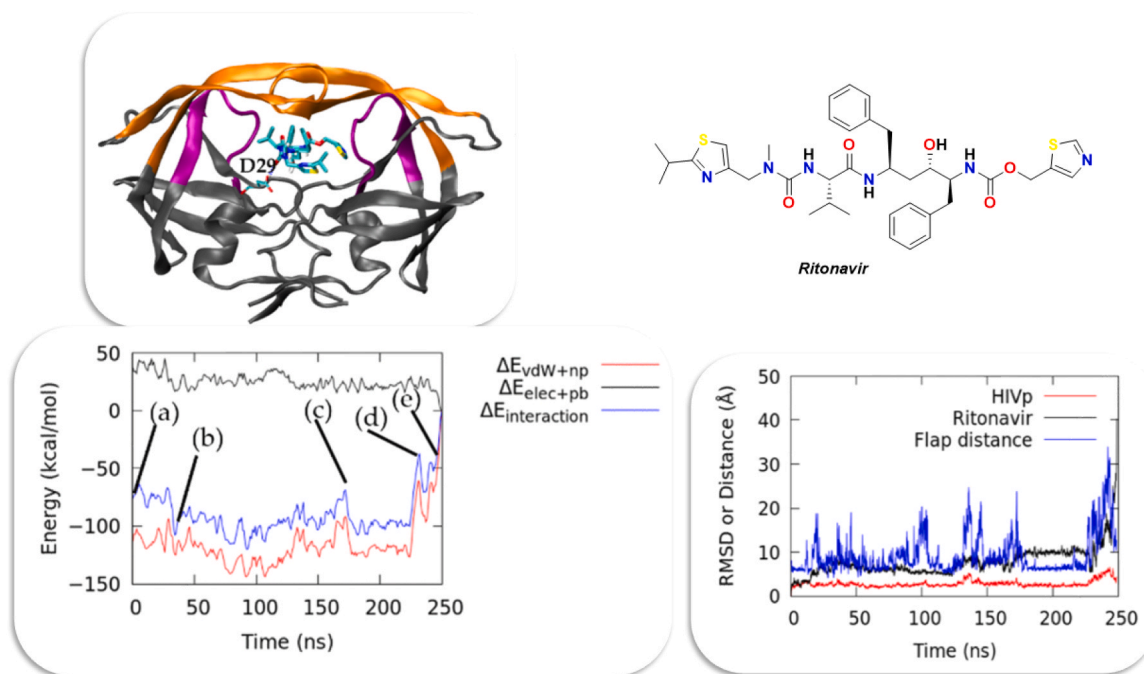


Fig. 7. Dynamics of binding and dissociation of ritonavir with HIV-1 protease (PDB ID: XK263) are shown here. This figure combines structure, chemical structure, binding free energy and RMSD results to show that ritonavir acts as a protease inhibitor as well as a CYP3A4 modulator.

effect increases the plasma concentration and half-life of these co-administered drugs, leading to a higher therapeutic benefit from the drugs when ritonavir is present.

To further review the binding behavior of Ritonavir, molecular dynamics (MD) simulations were conducted to analyze the interaction between it and the HIV-1 protease active site over time. These simulations showed that Ritonavir stabilizes its binding by virtue of both hydrophobic interactions and hydrogen bonding with critical residues of the protease, such as Asp25, Asp29, and Ile50. The first docking pose was consistently maintained throughout the simulations, except for some small conformational variations observed among the flexible loops of the protease. The thiazole and phenyl rings of Ritonavir made continuous van der Waals contacts with the hydrophobic subsites (S1, S2, S1', S2') between which the central hydroxyl group provided coordination for the catalytic aspartate dyad. Confirmation of the stability of the complex came from the analysis of RMSD and RMSF, which exhibited little deviation in audio chamber prints. These dynamic insights complement the static crystal structure data and provide a more comprehensive understanding of the mechanism of action of Ritonavir, one that affords it its slow dissociation rate and long residence time and underpins its efficacy as a dual antiviral and pharmacokinetic enhancer.

3.4.2.3. SAR of Ritonavir. The unique dual-action of ritonavir (protease inhibition and pharmacokinetic enhancement) comes from the intentional but careful design driven by SAR: Core Hydroxyethylamine Scaffolding: Like any other PIs, the hydroxyethylamine scaffold is important for transition-state mimicry and binding to the catalytic aspartates. Thiazole Rings: Both n-methyl and iso-propyl thiazole rings are important in occupying the protease hydrophobic subsites (S2/S2' and S1/S1'), respectively. The iso-propyl substituent on the thiazole ring was the SAR finding; the iso-propyl provided lipophilic interactions with the hydrophobic subsite that were superior to those from the smaller substituents, which increased binding affinity greatly. Pyridyl Group: The pyridyl group is unsubstituted and contributes to binding as well as is critical for mechanism-based inhibition of CYP3A4. In this case, heteroatom coordination (the nitrogen coordinates with the heme iron of cytochrome P450) creates a metabolite that inhibits CYP3A4. Terminal

Carbamate: Ritonavir contains the classical phenyloxycarbonyl-thiazolylmethyl carbamate. SAR indicates that the two thiazole rings (one each with a methyl and iso-propyl substituent) and the carbonyl attracted enough binding energy for high potency. The e and f groups (the terminal carbamate and carbonyl) are part of the molecule's metabolism and both contribute to ritonavir's self-inhibitory properties on CYP3A4.

3.4.2.4. Side effect. Ritonavir is a potent inhibitor of the CYP3A4 enzyme, resulting in extensive drug interactions that may elevate the risk of toxicity or diminish the therapeutic efficacy of co-administered medications. Given its potential for serious adverse effects such as loss of appetite, upper stomach pain (that may spread to the back), nausea, vomiting, dark urine, yellowing of the skin or eyes (jaundice), night sweats, swollen glands, cold sores, cough, wheezing, diarrhea, weight loss, trouble speaking or swallowing, problems with balance or eye movement, patients receiving ritonavir require careful monitoring, particularly those with preexisting medical conditions or those on complex medication regimens (Aarnoutse et al., 2003).

3.4.2.5. Indinavir. Indinavir, a protease inhibitor (PI), is a vital antiretroviral drug used in the treatment of HIV/AIDS. As one of the earliest drugs developed in the protease inhibitor class, Indinavir plays a crucial role in inhibiting HIV replication by targeting the HIV protease enzyme. Mechanistically, Indinavir binds to the active site of HIV protease, blocking its ability to cleave viral polyproteins into mature, functional proteins essential for viral replication. By preventing this critical step, Indinavir inhibits the formation of infectious viral particles, thereby reducing viral load and slowing disease progression. Its effectiveness in suppressing viral replication has made it a cornerstone in combination antiretroviral therapy (cART) for managing HIV infection (Plosker and Noble, 1999; Chavan et al., 2001).

3.4.2.6. Synthesis of Indinavir. Indinavir (4) is synthesized through a stepwise construction of a hydroxyethylene dipeptide isostere designed to mimic the transition state of the HIV-1 protease. The synthesis begins with the condensation of 1-((3aS,8aR)-2,2-dimethyl-8,8a-dihydro-2H-

indeno[1,2-*d*]oxazol-3(3*a*H)-yl)-3-phenylpropan-1-one (1) via the in-situ enolate generation using *n*BuLi which showed the electrophilicity character of alkoxy epoxide toward the nucleophilic attack then alkylation of enolate Li which can easily attack with tert-butyl ((1*R*)-1-(oxirane-2-yl)-2-phenylethyl)carbamate to afford the tert-butyl ((2*R*,3*S*)-5,6-dibenzyl-7-((3*a*S,8*a*R)-2,2-dimethyl-8,8a-dihydro-2*H*-indeno[1,2-*d*]oxazol-3(3*a*H)-yl)-3-hydroxy-7-oxo-1-phenylheptan-2-yl)carbamate (2), a hydroxyindane framework, to afford the key amide intermediate (3) in 75 % yield. The hydroxyethylene isostere is introduced stereoselectively, providing effective mimicry of the proteolytic cleavage site. Subsequent elaboration of the side chains, followed by deprotection and final coupling, leads to the formation of Indinavir (4) in 55 % yield. Across the multi-step route, the overall yield is approximately 15–25 %, which reflects the complexity of peptidomimetic drug synthesis and the necessity for protecting group strategies. This efficient route highlights the careful stereocontrol required for the development of potent HIV-1 protease inhibitors (Askin et al., 1994). (Scheme 6)

3.4.2.7. CADD investigation of Indinavir. Indinavir's primary target was identified as HIV protease, an enzyme required for viral maturation. Through X-ray crystallography (PDB ID: 2Q5K), the enzyme's structure was determined, revealing its active site and dimeric nature. Modeling of the enzyme's structure using different docking analyses to simulate how Indinavir would bind to the active site, blocking the cleavage of viral polyproteins and halting viral maturation (Okafor et al., 2022). The binding affinity of Indinavir to HIV protease was computationally predicted, offering valuable insights into its most energetically favorable binding modes and the strength of enzyme-ligand interactions (Fig. 8). These simulations provided a deeper understanding of how Indinavir interacts with the protease enzyme at the molecular level. Additionally, virtual screening was conducted to optimize potential drug candidates based on binding affinity, selectivity, and pharmacokinetic properties. Structure-Activity Relationship (SAR) analysis was employed to identify functional groups that significantly contributed to the drug's efficacy. Through targeted structural modifications, researchers enhanced binding affinity and potency, refining Indinavir's pharmacological profile. This iterative drug design process revealed key molecular features that

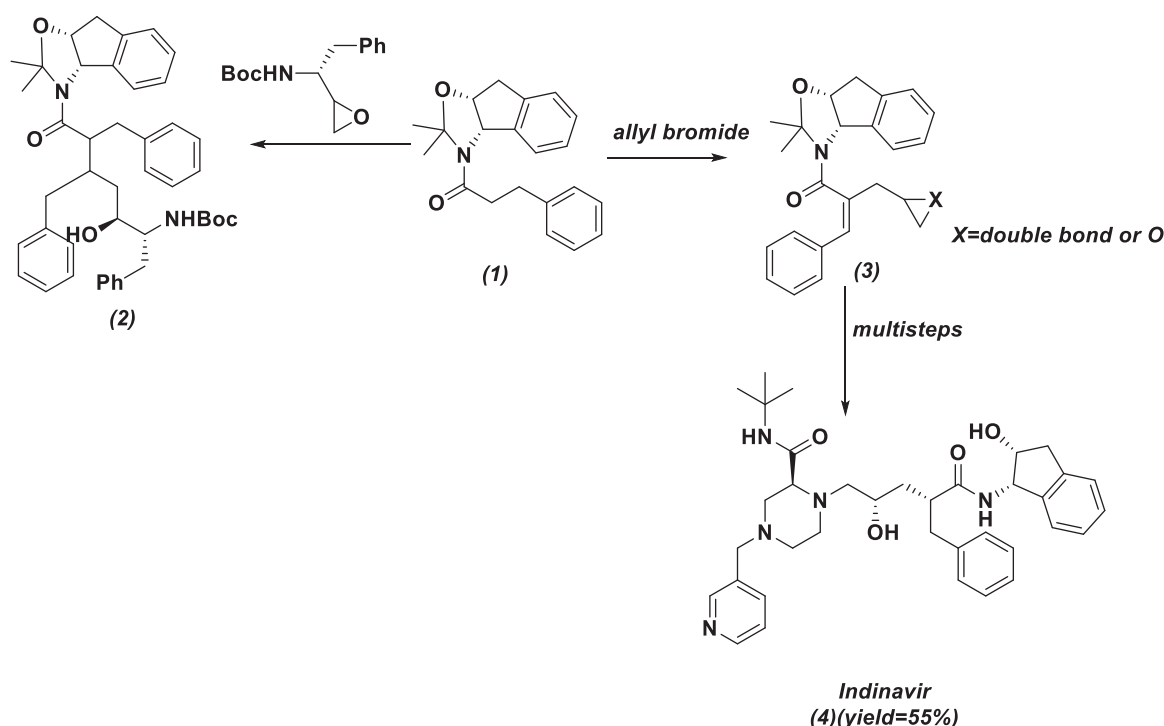
strengthened Indinavir's inhibitory effect, including hydrogen bonding, hydrophobic interactions, and electrostatic interactions within the enzyme's active site. Furthermore, in silico ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) modeling was applied to evaluate and optimize Indinavir's pharmacokinetics, improving its oral bioavailability and overall therapeutic potential.

3.4.2.8. Side effects. Patients prescribed indinavir should ensure adequate hydration to reduce the risk of kidney stone formation and undergo routine monitoring for metabolic imbalances and liver function abnormalities. Additionally, careful management of drug interactions is essential, as indinavir can significantly alter the metabolism of co-administered medications (Isaac et al., 2004; Rhame Frank et al., 2004).

3.4.3. Amprenavir

Amprenavir is a protease inhibitor (PI) used in the treatment of HIV/AIDS. As part of a drug class specifically designed to inhibit the HIV protease enzyme, Amprenavir effectively blocks the cleavage of viral polyproteins, a critical step in the HIV replication cycle. By preventing the formation of mature, infectious viral particles, it helps to reduce viral load, slow disease progression, and enhance patient outcomes (Burger and Abraham, 2003). Amprenavir binds to the HIV protease enzyme (PDBID: 3NU3), inhibiting its ability to cleave viral polyproteins into functional proteins essential for the formation of mature viral particles. By disrupting this critical step in the viral maturation process, Amprenavir effectively reduces viral load and slows the progression of HIV infection, thereby enhancing therapeutic outcomes (Sham et al., 1998). Amprenavir specifically targets the active site of the HIV protease enzyme, blocking its function and thereby preventing the virus from producing infectious viral particles. This impairs the virus's ability to replicate, lowering the viral load in the body and improving immune system function (Kim et al., 1995).

3.4.3.1. Synthesis of Amprenavir. The synthesis begins with ((2*S*,3*S*)-3-amino-2-hydroxy-4-phenylbutyl)-*N*-isobutyl-4-nitro benzene sulfonamide (1), which contains a sulfonamide backbone bearing a nitro-substituted aromatic ring and a secondary hydroxyl group. In the first



Scheme 6. Multistep reactions for the development of the Indinavir drug.

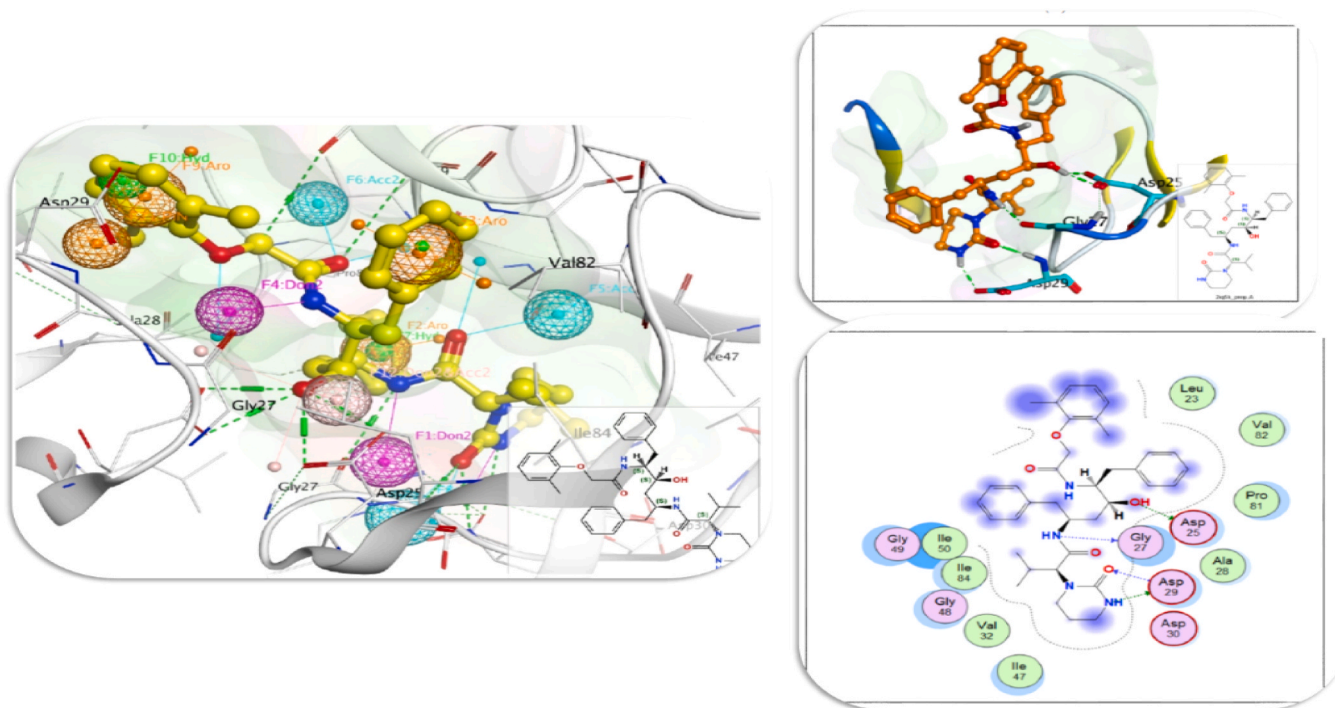
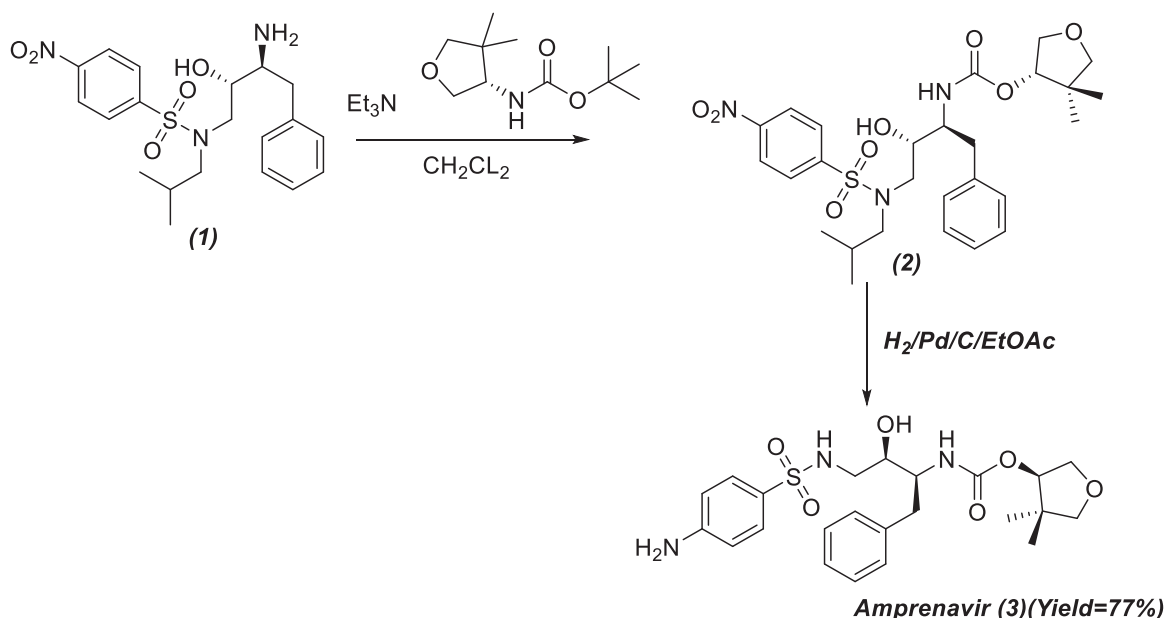


Fig. 8. Pharmacophore annotations of indinavir docked into HIV protease PDB ID: 2Q5K. Both models show hydrogen bonding and hydrophobic interactions that drive the inhibitor strength.

step, (1) is treated with (R)-4,4-dimethyltetrahydrofuran-3-yl (2,5-dioxopyrrolidin-1-yl) carbonate in presence of methylene chloride and drops of triethyl amine and stirred for 4 h then poured into brine and extracted with EtOAc to afford the (R)-4,4-Dimethyltetrahydrofuran-3-yl ((2S,3 R)-3-Hydroxy-4-((N-isobutyl-4-nitrophenyl) sulfon amido)-1-phenylbutan-2-yl)carbamate(2). This step activates the hydroxyl group and promotes carbamate bond formation, yielding the protected carbamate intermediate (2). The Boc group in this step plays a dual role: it temporarily protects the amine functionality and ensures selectivity by preventing unwanted side reactions. In the second step, the intermediate (2) undergoes hydrogenolysis ($H_2/Pd-C$ in MeOH), which serves to

remove the Boc protecting group and reduce the nitro group where necessary, furnishing the final Amprenavir drug (3)(yield=77 %). This step ensures that the free amine functionality is restored, which is essential for the drug's biological activity (Athawale et al., 2019) as showed in Scheme 7.

3.4.3.2. CADD studies of Amprenavir. The binding affinity of HIV-1 protease proteins, such as PDBID:3NU3 X-ray crystallography revealed the dimeric nature of HIV protease and its active site and identification of hydrophobic interactions, hydrogen bonding, and electrostatic interactions which determine the binding affinity of Amprenavir inside the



Scheme 7. Preparation of Amprenavir.

pocket of 3NU3, and investigation of the QSAR of it as displayed in Fig. 9 (Dubey and Gowtham, 2011). Furthermore, the interaction Amprenavir bound to HIV Protease (PDB: 3NU3) showed a sulfonamide moiety and a tetrahydrofuran (THF) heterocyclic ring, which are important pharmacophoric features and is a very strong hydrogen bond acceptor. It typically forms stronger hydrogen bonds with the backbone NH groups of catalytic water and the protease active site residues Asp29/Asp30, which can help immobilize amprenavir, also tetrahydrofuran (THF) moiety which make Amprenavir is optimized to fit in the S2 subsite of protease and the THF ring engages in favorable van der Waals contact with various hydrophobic residues that line this S2 pocket, then Transition-state mimicry: Similar to other PIs, the hydroxyl group of amprenavir interacts with the catalytic aspartate dyad (Asp25/Asp25'), mimicking the transition-state. So Amprenavir is a unique shaped inhibitor and functional groups induce movement in the protease during binding (e.g. some minimal "flap" movement), while still optimizing fit to the active site, which also helps to create a stronger binding affinity as displayed in Fig. 9.

So it is concluded that these initial-generation PIs share a common theme of fighting against viral entropy. HIV protease is intrinsically flexible, and its substrates are polymorphic. The drugs were designed to fit a static crystal structure, so their clinical effectiveness is a function of how well they bind across all of the conformational states the protease can adopt. Saquinavir and indinavir are more rigid inhibitors and can be disrupted easily because of their strict steric requirements related to the altered volume and dynamics of the subsites (S2/S2' for saquinavir; S1' for indinavir). Ritonavir and Amprenavir, because they were bulkier and more complex structures, used a larger surface area. However, with the larger surface area comes a larger 'point of contact' that can also potentially be disrupted. The reason ritonavir is such a good booster was ironically also the basis for its eventual susceptibility as an antiviral when the virus mutated to bypass its rigid grasp. The development of resistance is a direct result of the virus investigating the dynamic conformational landscape of the protease to find states where substrate processing is still possible while inhibitor binding is not. The understanding of these dynamic shifts has been the basis for developing next-generation PIs like darunavir, which have utilized cleverly placed bonds to ensure stronger and more stable interactions with the protease backbone, making it less responsive even when changes occur in the dynamics of the sidechains.

3.4.3.3. Side effects. Patients receiving amprenavir should undergo regular monitoring for liver function abnormalities, metabolic imbalances, and dermatological reactions. Given its potential for significant drug interactions, meticulous management is essential when prescribing amprenavir alongside other medications (Arvieux and Tribut, 2005).

4. Influenza

Influenza is a contagious respiratory illness caused by the influenza

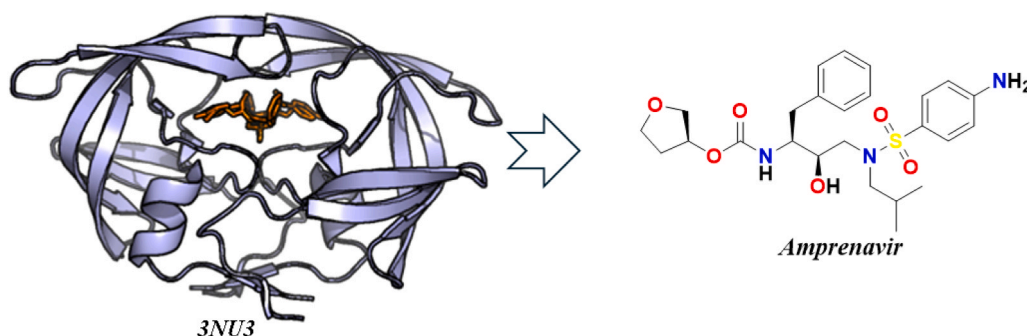


Fig. 9. The docking pose of amprenavir within HIV protease active site PDB ID: 3NU3 shows the structural alignment of the sulfonamide group reactions with the catalytic residue diagrams and the stereoelectronic fit of the sulfonamide that drives inhibition.

virus. It mainly affects the throat, nose, and lungs, causing symptoms like cough, fever, sore throat, fatigue, muscle aches, and chills. Whereas most cases are mild, influenza can result in severe complications, particularly in high-risk groups, including young children, pregnant women, the elderly, and individuals with weakened immune systems or pre-existing health conditions, and it is divided into Influenza A, B, C, and D as displayed in Fig. 10 (Bahadur Gurung et al., 2020; Javanian et al., 2021; Chan et al., 2021).

4.1. Oseltamivir (Tamiflu)

Oseltamivir, also known as Tamiflu, is an antiviral medication that is widely used to treat and prevent influenza A and B infections. It belongs to the neuraminidase inhibitor class of antiviral drugs, which function by targeting the viral enzyme neuraminidase a key factor in viral replication and transmission. By inhibiting this enzyme, Oseltamivir effectively curtails the spread of the virus to healthy cells, reducing infection severity and progression. Oseltamivir acts by blocking neuraminidase activity, an enzyme necessary for the release of newly formed viral particles from infected host cells. By disrupting this process, the drug prevents viral replication and limits the spread of infection, thereby alleviating flu symptoms and shortening the duration of illness (Russell et al., 2006).

4.2. Synthesis of Tamiflu

The synthetic route toward **Tamiflu (7)** begins with pure (1S,2S,6S)-

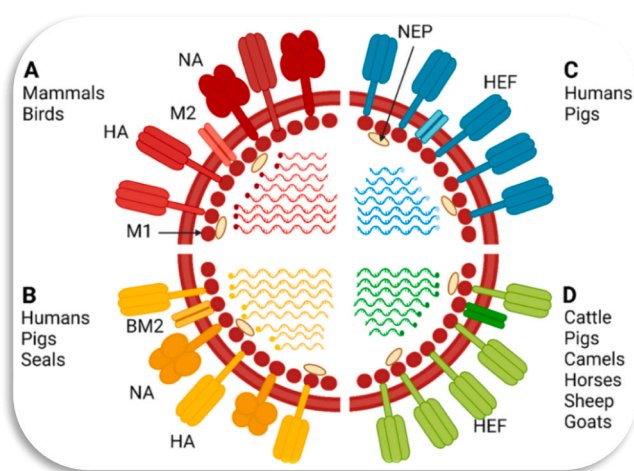


Fig. 10. Classification of influenza viruses into A, B, C, and D, and the respective segmentation of their genomes. Influenza A and B share eight RNA segments, whereas C and D contain seven segments, which confer genetic diversity and are important to research aimed at antiviral drug development.

7-oxabicyclo[4.1.0]hept-3-en-2-yl 4-nitrobenzoate (**1**) (Brandsma et al., 1990). The subsequent ring opening of epoxide facilitated by cesium chloride under NaN_3 and MeCN conditions provided azido alcohol (1*S*, 5*R*, 6*S*)-5-azido-6-hydroxycyclohex-2-en-1-yl 4-nitrobenzoate (**2**) which convert to aziridines, Regioselective ring opening of **3** was achieved using tetramethyl guanidinium azide (TMGA), affording compound (1*S*, 5*R*, 6*R*)-6-acetamido-5-azidocyclohex-2-en-1-yl 4-nitrobenzoate (**4**). Reduction of azide **4**, followed by N-Boc protection, led to diamide **5** with an. Subsequent base hydrolysis of compound (**5**), followed by oxidation under Dess–Martin periodinane conditions, furnished enantiopure Shibasaki's intermediate (**6**). Also, the HPLC analysis confirmed the formation of single stereoisomer tert-butyl ((1*R*, 6*R*)-6-acetamido-5-oxocyclohex-3-en-1-yl)carbamate (**6**) (Li, 2009). Notably, Shibasaki's intermediate **6** serves as a key precursor and can be readily transformed into Tamiflu in seven steps as displayed in Scheme 8 (Gunasekera, 2012). The final coupling of either (**6a**) or (**6b**) with the phosphonomethoxypropyl amine gave the target drug Tamiflu (**7**) in 70 % yield. Overall, the multistep synthesis proceeds in an isolated yield of approximately 20–35 %, consistent with literature reports for this class of nucleotide analogues.

4.3. CADD action of Tamiflu(Oseltamivir)

The high-resolution X-ray crystallography studies of influenza neuraminidase (PDB ID: 2HU4) (Russell et al., 2006) provided a comprehensive structural framework of the enzyme's active site, offering critical insights into the molecular interactions necessary for effective inhibition as displayed in Fig. 11. This structural data was critical in guiding the rational design of Oseltamivir, ensuring its strong affinity for the neuraminidase binding pocket. Computational molecular docking simulations validated the optimal binding of Oseltamivir within the neuraminidase active site, precisely targeting key amino acid residues such as Arg152, Arg292, and Glu119 (Amaro et al., 2011). The interaction profile was further refined through the optimization of hydrophobic interactions and hydrogen bonding, enhancing Oseltamivir's binding stability, efficacy, and inhibitory potential against the influenza virus (Yurtdaş-Kurumluoğlu et al., 2022).

The binding of Oseltamivir involves a dramatic conformational state change in the neuraminidase active site, namely the 150-loop which is

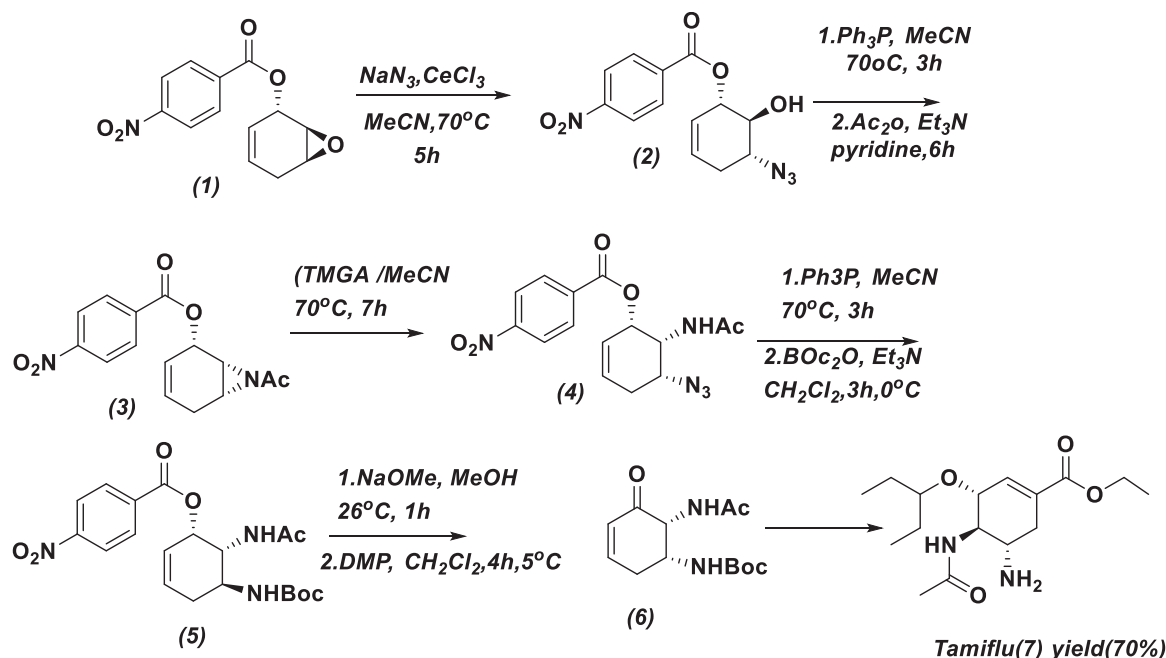
flexible. MD simulations have shown that Oseltamivir binding induces a series of changes that include a closed conformation of this flexible 150-loop, which produces a hydrophobic pocket into which the pentyl ether side chain of Oseltamivir can fit. This '150-cavity' does not exist in the apo-enzyme or the enzyme with its natural substrate, sialic acid, bound. The ability to induce this conformational change is a major driver for specificity and potency of Oseltamivir, and resistance mutations generally stabilize the open form of the 150-loop (e.g., H274Y), thus preventing the induced-fit mechanism and therefore reducing binding.

4.4. SAR of oseltamivir

The design of oseltamivir (a carbocyclic transition-state analogue of sialic acid) was optimized through SAR to further improve the already increased oral bioavailability of its predecessor, zanamivir. Cyclohexene Core: The carbocyclic ring acts as a stable scaffold to mimic the stable pyranose ring of sialic acid. The double bond gives the structure planarity which provides the correct conformation to allow direct binding. 3-Pentyl Ether Side-chain: This was the biggest SAR advancement for oseltamivir, replacing zanamivir's polar glycerol side chain with this lipophilic ether gave the greatest boost to oral bioavailability through greater membrane permeability. The 3-pentyl ether side chain induced the formation of the so-called "150-cavity" in neuraminidase resulting in unique hydrophobic pocket formation which would enhance binding. Amino Group: The free amino group present at the C5 position is involved in a key salt bridge with the conserved Glu119 residue located in the active site. Masking this group by acetylation (as in the prodrug oseltamivir phosphate) facilitates its absorption in vivo before hydrolysis to the active form. Carboxylate Group: The negatively charged carboxylate interacts with three conserved arginine residues (Arg118, Arg292, Arg371) in the active site. Esterification of this group is prohibited because it disrupts such crucial ionic interactions.

4.5. Side effects

Neuropsychiatric Effects: These effects are more commonly reported in children and adolescents and, in rare cases, may lead to self-harm (Antipov and Pokryshevskaya, 2019). Severe Allergic Reactions: In rare instances, severe dermatological conditions such as



Scheme 8. Steps for the Synthesis of Tamiflu.

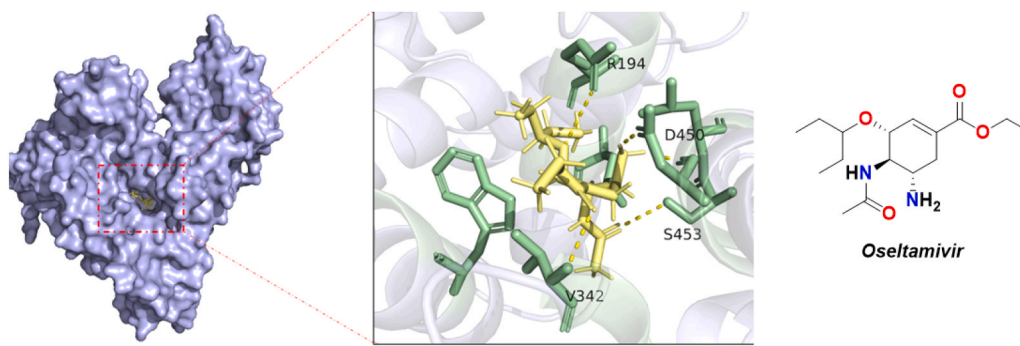


Fig. 11. The molecular docking of oseltamivir (Tamiflu) to the binding pocket of the influenza neuraminidase (PDB ID: 2HU4). Biological interactions consisted of hydrogen bonding with Arg152, Arg292, and Glu119, and stabilization through conformational changes made with respect to the 150-loop.

Stevens-Johnson syndrome and toxic epidermal necrolysis may occur. Hepatic Complications: Oseltamivir may cause liver dysfunction, manifesting as jaundice (yellowing of the skin or eyes), dark-colored urine, or persistent nausea, which may indicate underlying hepatic impairment (Cohen, 2014).

4.6. Coronavirus (SARS-CoV-2)

SARS-CoV-2 is an enclosed, single-stranded RNA virus with spike (S) proteins that aid viral entrance by attaching to host cells' angiotensin-converting enzyme 2 (ACE2) receptors, as illustrated in Fig. 12. The virus spreads mostly by respiratory droplets produced by an infected person coughing, sneezing, or speaking. Indirect transmission can also occur by contact with contaminated surfaces (fomites) followed by hand-to-face contact with the mouth, nose, or eyes (Fehr and Perlman, 2015; Masters, 2006; Race et al., 2020).

4.7. Nirmatrelvir/ritonavir (Paxlovid)

Nirmatrelvir/ritonavir (Paxlovid) inhibits a cysteine residue in the 3C-like protease (3CLPRO) of SARS-CoV-2 (Jakharia et al., 2022). This cysteine is responsible for the action of the 3CLPRO of SARS-CoV-2 and maybe other members of the coronavirus family (Focosi et al., 2023), and binds to the protease active site, inhibiting the enzyme's activity. This inhibition prevents the cleavage of viral polyproteins, leading to the formation of immature, non-infectious viral particles. Protease inhibitors are typically administered in combination with at least two other antiretroviral drugs to enhance therapeutic efficacy and reduce the

risk of drug resistance (Bartoletti et al., 2022; Li et al., 2023).

4.8. Synthesis of Paxlovid

This scheme shows the multistep synthesis of **Nirmatrelvir (10)**, the active antiviral agent in Paxlovid. The synthetic sequence proceeds through several intermediates, some of which are not isolated, and the overall efficiency is reported in the literature (Kincaid et al., 2022a). The transformation of N-Boc-protected t-leucine (1) began with its conversion into the thioester derivative (2) using di-2-pyridyldithiocarbonate DPDTTC (2) in the presence of EtOAc and catalytic DMAP at room temperature, affords intermediate (3). Subsequent addition of bicyclic proline (4) ~80–85 % yield in the presence of N-methylmorpholine (NMM) and ethyl acetate (EtOAc) led to the formation of compound 5, which was not isolated. This intermediate was then reduced with LiOH, yielding (1 R,2S,5S)-3-((S)-2-((tert-butoxycarbonyl)amino)-3,3-dimethylbutanoyl)-6,6-dimethyl-3-azabicyclo[3.1.0]hexane-2-carboxylic acid (6) (Kincaid et al., 2022b). In parallel, compound (8) is treated with EtOAc/NMM to give ((S)-1-((1 R,2S,5S)-2-(((S)-1-cyano-2-((S)-2-oxopyrrolidin-3-yl)ethyl)carbamoyl)-6,6-dimethyl-3-azabicyclo[3.1.0]hexan-3-yl)-3,3-dimethyl-1-oxobutan-2-yl)carbamate (9) in ~75–80 % yield. Both intermediates converge to generate (7), also not isolated, which undergoes coupling and Boc deprotection/reprotection steps. The final deprotection underwent azeotropic drying using recoverable toluene. Finally, N-Boc deprotection was performed using a concentrated solution of HCl/dioxane in CH₃CN, yielding the target compound Nirmatrelvir(10) in an isolated yield of 67 % (Kincaid et al., 2022a), as outlined in Scheme 9.

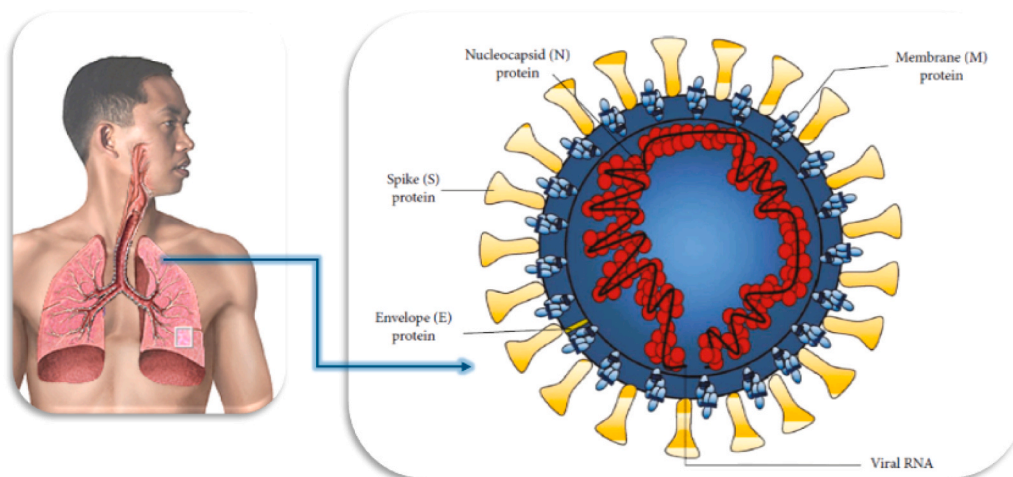


Fig. 12. Morphological anatomy of SARS-CoV2, containing the viral genome and spike glycoproteins that allow binding to the ACE2 receptor to facilitate entry into host cells. It also presents pathways for respiratory tract infection.

4.9. CADD of Nirmatrelvir/ritonavir (Paxlovid) and its SAR

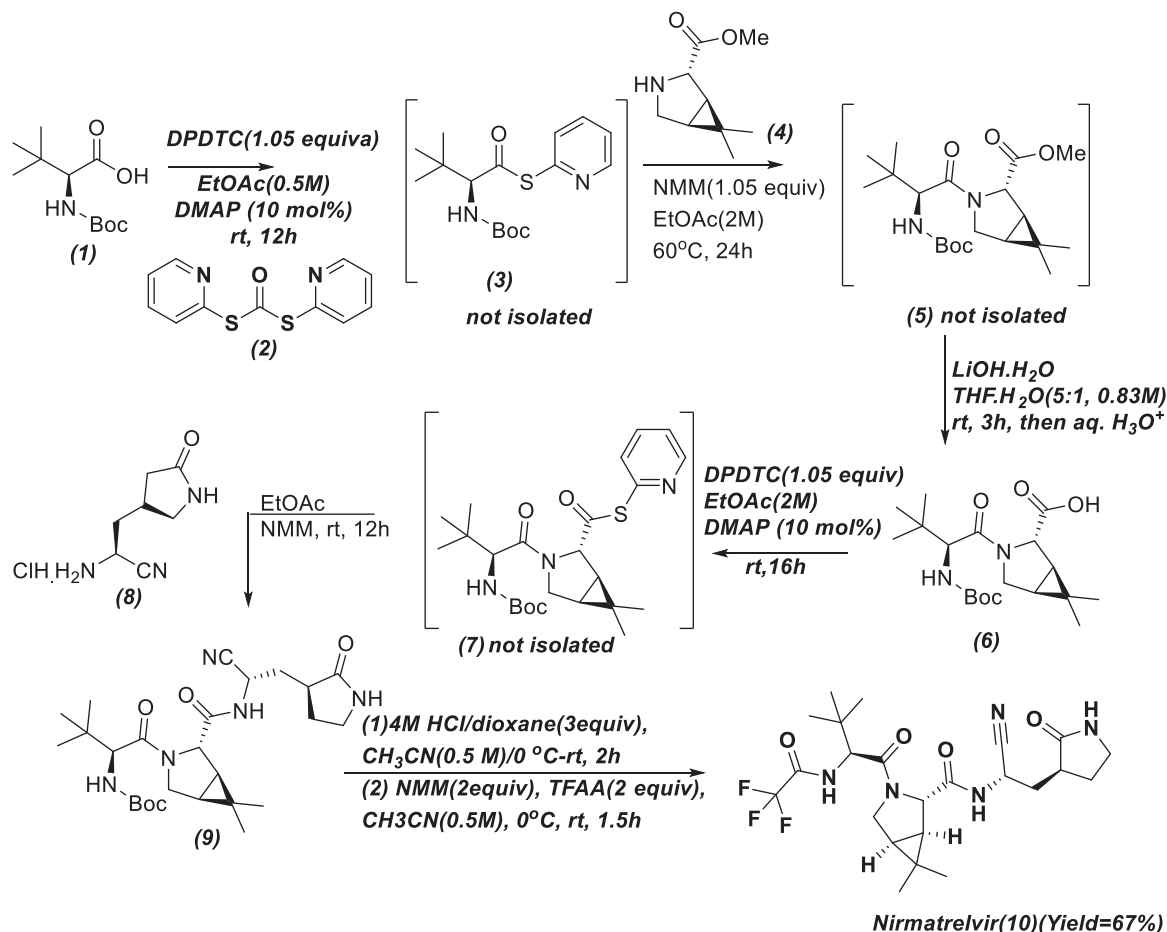
Nirmatrelvir/ritonavir (Paxlovid), a protease inhibitor created utilizing SBDD, targets the SARS-CoV-2 major protease (Mpro). Computational docking and dynamics simulations were used to aid compound optimization. Among 1072,004 non-hospitalized COVID-19 patients in Hong Kong during BA.2.2 wave, 5663 who received N/R had a lower mortality and hospitalization risk, regardless of vaccination status and age (Wong et al., 2022). During the Omicron wave, patients prescribed N/R in Massachusetts and New Hampshire (87 % vaccinated) had a 45 % lower risk of hospitalization within 14 days than those who did not take the protease inhibitor. Fig. 13 (Shekhar et al., 2022; Makhloufi et al., 2023). The interaction Nirmatrelvir (PF-07321332) bound to SARS-CoV-2 Main Protease (Mpro)

showed the active drug in Paxlovid, bound to the SARS-CoV-2 main protease (Mpro or 3CLpro). Nirmatrelvir is a covalent and reversible inhibitor. Through covalent binding interaction is between the inhibitor's nitrile group ($-C\equiv N$) and the catalytic cysteine (Cys145) of Mpro. The nitrile acts as an electrophilic "warhead" and forms a reversible, covalent thioimide adduct with the thiol group of Cys145 that directly obstructs the enzyme's catalytic machinery. Non-Covalent Recognition, which is stabilized by the presence of several non-covalent interactions before covalent bond formation. Firstly, with hydrogen bonding, the lactam carbonyl oxygen forms an essential hydrogen bond with the backbone NH of Cys145. The tertiary-butyl group fits in the S2 pocket, and the trifluoroacetamide group engages the S4 pocket. Then, hydrophobic Interactions with the bicyclic proline group (3-azabicyclo [3.1.0]hexane) scaffolding are rigid and allow for optimal positioning of the warhead while filling the S1 and S3' pockets, resulting in high

selectivity and potency. So, stabilizing the Dimer: The simulations also indicate that nirmatrelvir binding could stabilize the functional dimeric state of the Mpro, which is the functional form of Mpro. By preventing the viral polyprotein processing as well as its replication by locking the enzyme in an inhibited state, as demonstrated in Fig. 13.

Its SAR of the drug with a pocket of protein through Nitrile Warhead ($-C\equiv N$): This electrophile is required for covalent, reversible inhibition as it forms a thioimide adduct with catalytic Cys145. SAR analysis showed other warheads (preferably aldehydes) performed either poorly in their selectivity or in their pharmacokinetic properties. Bicyclic Proline (3-azabicyclo[3.1.0]hexane): The structure is distinctive, rigid, and a critical SAR discovery. It optimally orients the warhead for attack on Cys145, whilst the bicyclic proline scaffold fills the S1 and S3 pockets. More flexible proline analogs demonstrated drastically lower potency. tert-Butyl Group (S2 Pocket): The (S)-tert-leucine-derived group is perfectly sized to fill the hydrophobic S2 pocket. However, smaller sizes (isopropyl) did not fill the size of the space efficiently, leading to weaker binding. Trifluoroacetamide (S4 Pocket): The $-CF_3$ group provides lipophilic interactions to the S4 pocket. The amide carbonyl also forms a vital hydrogen bond with the backbone NH of Glu166, which is very important for dimerization and catalytic activity.

Nirmatrelvir is a covalent inhibitor that binds reversibly to the catalytic Cys145 of its Mpro, SARS-CoV-2. MD simulations of the binding are necessary to examine the pre-covalent recognition complex of the drug. Simulations establish that the nitrile warhead must be perfectly oriented for nucleophilic attack with Cys145. The Rigid bicyclic proline moiety of Nirmatrelvir limits the flexibility of the drug and results in the most favourable spatial pre-reaction conformational orientation. The simulations also demonstrate how the inhibitor confines and stabilizes



Scheme 9. Multistep reaction for synthesis of Nirmatrelvir.

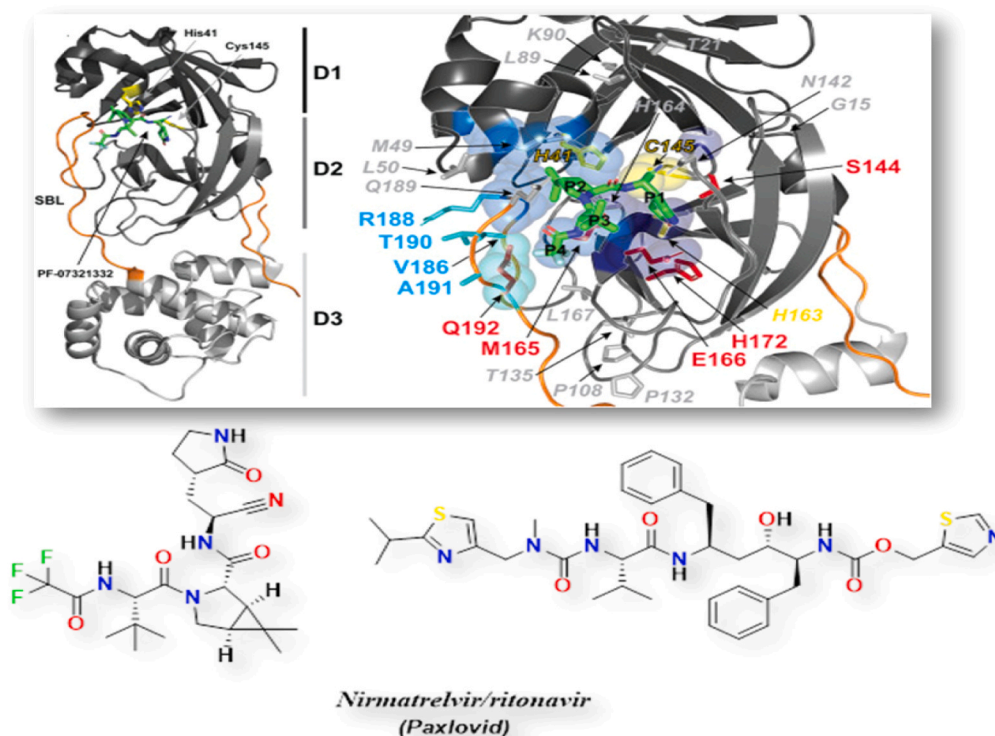


Fig. 13. Structure of SARS-CoV-2 main protease (Mpro) bound to the inhibitor nirmatrelvir (PF-07321332). The cartoon shows the covalent reversible interaction with Cys145, as well as the induced stabilization of the Mpro dimer conformational ensemble.

the dimeric form of Mpro, which is critical to its activity. Analysis of the covalent complex dynamics is also important to rationalizing the half-life of the bond and the residence time of the drug, which are critical for drug effectiveness.

4.10. Side effects

This drug may cause liver dysfunction, characterized by elevated liver enzyme levels, jaundice (yellowing of the skin and eyes), dark-colored urine, and persistent nausea. Patients with pre-existing liver conditions should be closely monitored. Trigger serious allergic reactions, presenting as skin rash, intense itching, facial, tongue, or throat swelling, and respiratory distress.

4.11. FDA-approved drugs

The current study examined a library of 75 antiviral medications licensed by the FDA, as well as 263 phytochemicals from various families such as prenol lipids, flavonoids, indoles and their derivatives, alkaloids, lignans, and organooxygen compounds. FDA-approved drugs have already undergone extensive preclinical and clinical trials verifying their safety in people; hence, they were not re-evaluated using in silico drug-like filters. To reduce medication attrition, plant-derived compounds were submitted to a virtual screening process, despite their generally safer profiles and reduced incidence of side effects. The oral bioavailability of the selected compounds was assessed using Veber's rule and the Rule of Five (ROF), which establishes that a compound is considered orally bioactive if its physicochemical properties fall within defined safety parameters, and there are some clinical drugs and their approval as shown in Table 1 (Bahadur Gurung et al., 2020).

5. Enhancing the drug discovery arsenal: repurposing and virtual screening

The drug discovery pipeline has a poor reputation for being slow,

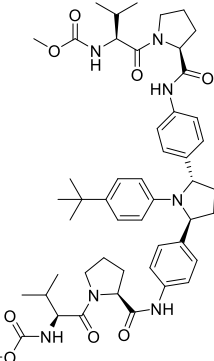
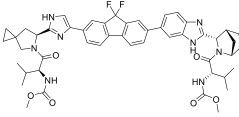
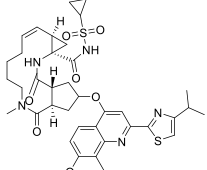
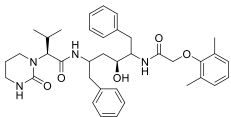
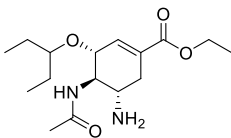
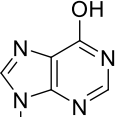
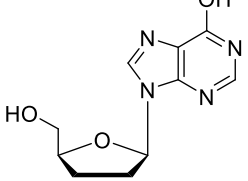
costly, and high-risk. These concerns have been highlighted recently in relation to important global health threats, including the COVID-19 pandemic, and thus innovative computational approaches (e.g. drug repurposing and virtual screening) have been utilized as a rapid way to identify existing drugs and novel compounds for therapeutic use (Pushpakom et al., 2019; Zhou et al., 2020). These computational approaches employ large databases of approved drugs (for repurposing) or chemical libraries (for virtual screening) to identify molecules that can bind to and modulate a novel target of interest.

5.1. Drug repurposing

Drug repurposing (or repositioning) is the study of existing, approved drugs that have potential new therapeutic indications. This has many advantages, including: known safety and pharmacokinetic profile in humans, lower development costs, and a faster pathway to study in humans (Ashburn and Thor, 2004). One good example mentioned throughout this review is ritonavir. Initially developed from scratch as an HIV-1 protease inhibitor, it was repurposed as a pharmacokinetic booster (to enhance the pharmacokinetic profile of antivirals for HIV) and then, most recently, in the COVID-19 combination therapy of Paxlovid (nirmatrelvir/ritonavir) to boost nirmatrelvir (Oprea and Overington, 2015; Beck et al., 2020). In addition to the aforementioned cases, large-scale efforts to repurpose drugs also emerged during the SARS-CoV-2 pandemic. In silico data mining/docking and molecular dynamics studies identified several FDA-approved drugs as possible agents for COVID-19. For example, chloroquine/hydroxychloroquine (an antimalarial) was suggested to have effectiveness against COVID-19 because it was expected to inhibit viral entry to cells via endosomal disruption. Baricitinib (a JAK inhibitor used to treat rheumatoid arthritis) was proposed to act to reduce cytokine storm and prevent viral endocytosis (Richardson et al., 2020; Lionta et al., 2014). There were mixed clinical outcomes for some of these candidates, but these activities demonstrate the effectiveness of computational hypotheses for quickly generating solutions during a crisis.

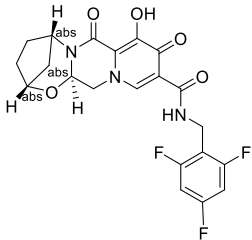
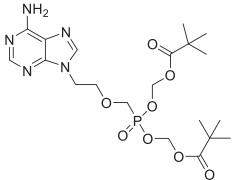
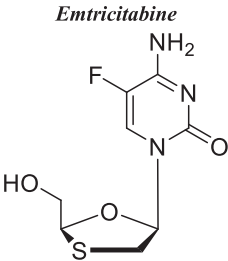
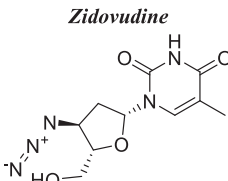
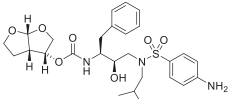
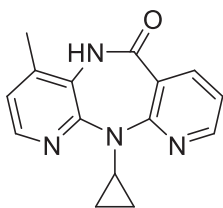
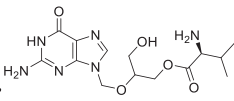
Table 2

List of FDA-approved clinical medicines chosen for molecular docking investigations.

Drugs	Drug Bank ID	Therapy	Approved in Year
Ombitasvir 	DB09296	Chronic Hepatitis C	2014 (Feld et al., 2014)
Ledipasvir 	DB09027	Chronic Hepatitis C	2014 (Fahim et al., 2021)
Simeprevir 	DB06290	Chronic hepatitis C virus	2018 (Sanford, 2015)
Simeprevir Lopinavir 	DB01601	Human immunodeficiencyvirustype1 (HIV–1) infection.	2000 (Fischer and Ganellin, 2010)
Oseltamivir 	DB00198	Influenza viruses A and B infections	1995 (Ritesh et al., 2010)
Oseltamivir Baloxavirmarboxil 	DB13997 DB00900	Influenza A and Influenza B infections HIV infection	2019 (O'Hanlon and Shaw, 2019) 1991 (Perry and Noble, 1999)
Didanosine 	DB11799	HIV–1andHIV–2infection	2018 (Markham, 2018 ; Hidalgo-Tenorio et al., 2024)

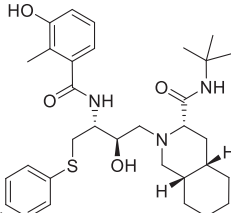
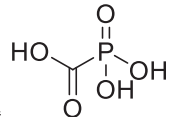
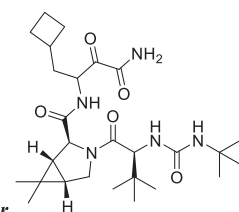
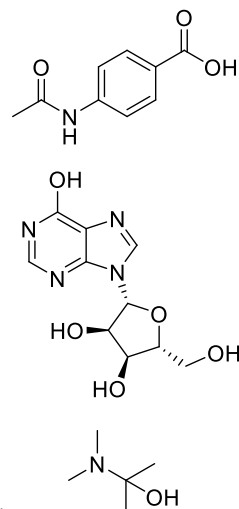
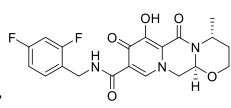
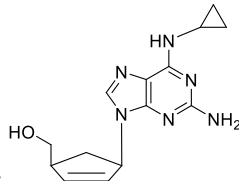
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Drugs	Drug Bank ID	Therapy	Approved in Year
 Bictegrovir Adefovir dipivoxil	DB00718	Hepatitis B	2002 (Qaqish et al., 2003)
 Adefovir dipivoxil Emtricitabine	DB00879	HIV-1 infection	2006 (Gupta and STDs, 2008)
 Emtricitabine Zidovudine	DB00495	HIV infection	1987 (Kinch and Patridge, 2014)
 Zidovudine Darunavir	DB01264	HIV-1 infection	2006 (Rittweger and Arastéh, 2007)
 Darunavir Nevirapine	DB00238	HIV-1 infection and AIDS.	1996 (Bowersox, 1996)
 Nevirapine Valganciclovir	DB01610	Cytomegalovirus infections	2001 (Kaye, 2001)
 Valganciclovir			

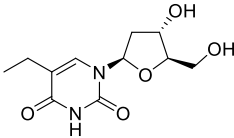
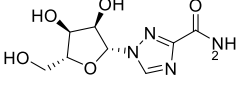
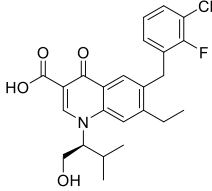
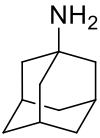
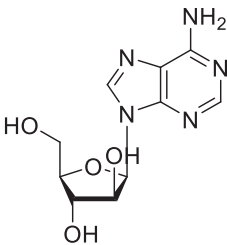
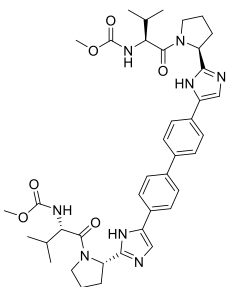
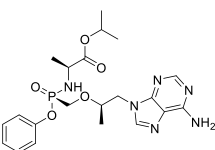
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Drugs	Drug Bank ID	Therapy	Approved in Year
Nelfinavir	DB00220	HIV infection	1997 (Jarvis and Faulds, 1998)
			
Foscarnet	DB00529	cytomegalovirus retinitis, HIV infection	1991 (Shankar Upadhyayula et al.)
			
Boceprevir	DB08873	Chronic Hepatitis C	2011 (Lin and Chung, 2014)
			
Inosine pranobex	DB13156	Viral infection	1971 (Nagpal et al., 2020)
			
Dolutegravir	DB08930	HIV-1 infection	2013 (Ballantyne and Perry, 2013)
			
Abacavir	DB01048	HIV infection and AIDS.	1998 (Cameron and Trubiano, 2017)
			

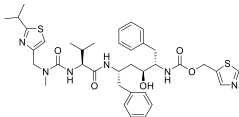
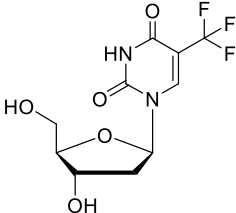
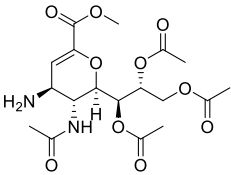
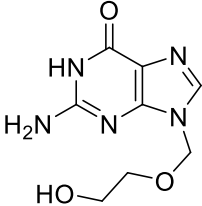
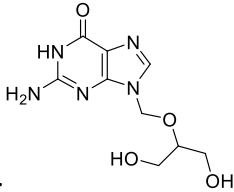
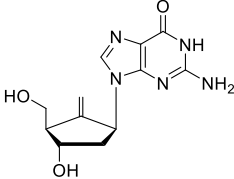
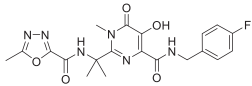
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Drugs	Drug Bank ID	Therapy	Approved in Year
Edoxudine 	DB13421	Herpes simplex virus type 1 and 2 infection	1998 (Hamuy and Berman, 1998)
Ribavirin 	DB00811	Hepatitis C and viral hemorrhagic fevers	1986 (Fischer and Ganellin, 2010)
Elvitegravir 	DB09101	HIV-1 infection	2015 (Sax et al., 2012)
Amantadine 	DB00915	Influenza A infection	2020 (Danysz et al., 2021)
Vidarabine 	DB00194	Herpesviruses, the vaccinia virus, and varicella-zoster virus infection	2004 (Kijjoa and Sawangwong, 2004)
Daclatasvir 	DB09102	Hepatitis C Virus (HCV) infection	2015 (McCormack, 2015)
Tenofovir alafenamide 	DB09299	Chronic hepatitis B and HIV-1 infection	2016 (Wassner et al., 2020)

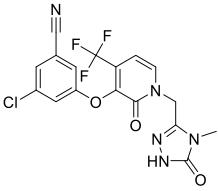
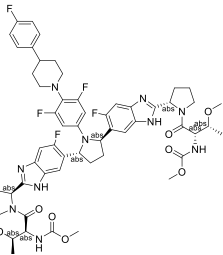
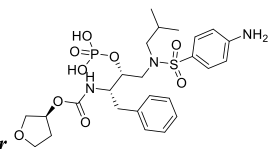
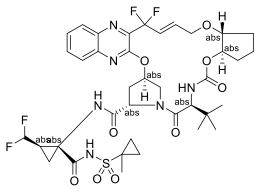
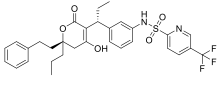
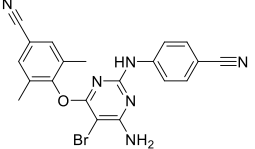
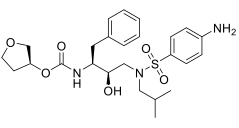
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Drugs	Drug Bank ID	Therapy	Approved in Year
Ritonavir 	DB00503	HIV infection	2020 (Dalva-Aydemir et al., 2015)
Trifluridine 	DB00432	Keratoconjunctivitis and recurrent epithelial keratitis	1980 (Patel et al., 2014)
Zanamivir 	DB00558	Influenza A and B virus infection	1999 (Tompa et al., 2021)
Acyclovir 	DB00787	Herpes simplex, Varicella-zoster, herpes zoster infection	1981 (Fischer and Ganellin, 2010)
Ganciclovir 	DB01004	AIDS-associated cytomegaly virus infections.	1988 (Fischer and Ganellin, 2010)
Entecavir 	DB00442	Hepatitis B infection	2005 (Rivkin and opinion, 2005)
Raltegravir 	DB06817	HIV infection	2007 (Deeks et al., 2008)

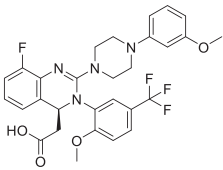
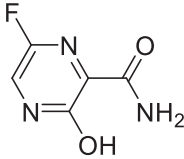
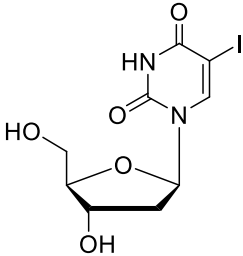
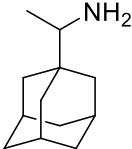
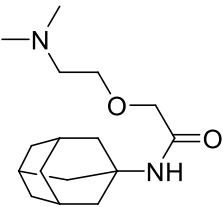
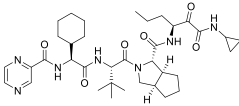
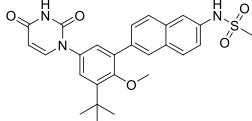
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Table 2 (continued)

Drugs	Drug Bank ID	Therapy	Approved in Year
Doravirine	DB12301	HIV – 1 Infection	2018 (Deeks, 2018)
			
Doravirine			
Pibrentasvir	DB13878	Hepatitis C virus(HCV)infection	2017 (Lamb, 2017)
			
Pibrentasvir			
Fosamprenavir	DB01319	HIV infection	2004 (Wire et al., 2006)
			
Fosamprenavir			
Glecaprevir	DB13879	HCV infection	2017 (Jonas et al., 2020)
			
Glecaprevir			
Tipranavir	DB00932	HIV infection	2008 (Doyon et al., 2005)
			
Tipranavir			
Etravirine	DB06414	HIV – 1 infection	2007 (Stellbrink, 2007)
			
Etravirine			
Amprenavir	DB00701	HIV infection.	1992 (Fischer and Ganellin, 2010)
			
Amprenavir			
Letermovir	DB12070	Cytomegalovirus(CMV)infection	2018 (El Helou and Razonable, 2019)

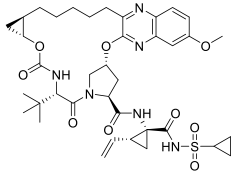
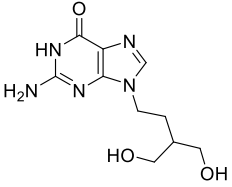
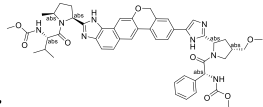
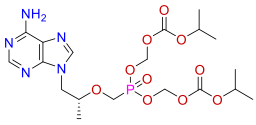
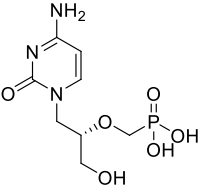
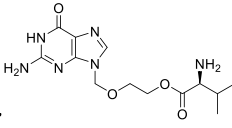
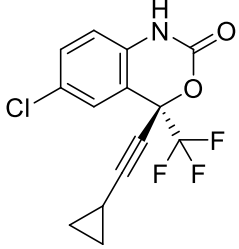
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Drugs	Drug Bank ID	Therapy	Approved in Year
 Letermovir Favipiravir	DB12466	Influenza	2019 (Pilkington et al., 2020)
 Favipiravir Idoxuridine	DB00249	Herpes simplex virus (HSV) infection	1962 (Wilhelmus, 2015)
 Idoxuridine Rimantadine	DB00478	Influenza.	2009 (Vorobjev, 2021)
 Rimantadine Tromantadine	DB13288	Herpes zoster and simplex virus infection	1989 (Ostheimer et al., 1989)
 Tromantadine Telaprevir	DB05521	Chronic Hepatitis C	2011 (Jacobson et al., 2011)
 Telaprevir Dasabuvir	DB09183	Chronic Hepatitis C	2014 (Trivella et al., 2015)
 Dasabuvir Grazoprevir	DB11575	Chronic Hepatitis C	2016 (Guo et al., 2017)

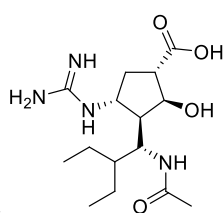
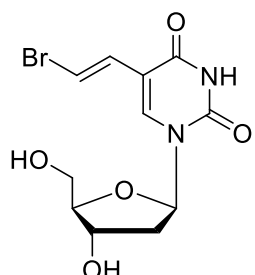
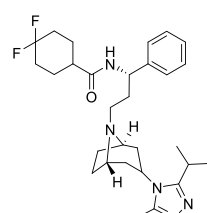
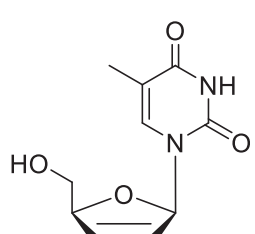
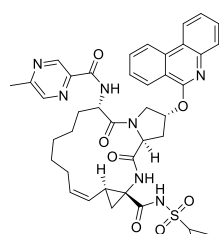
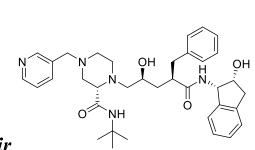
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Drugs	Drug Bank ID	Therapy	Approved in Year
 Grazoprevir			
Docosanol	DB00632	HSV infection	2000 (Katz et al., 1991)
 Docosanol			
Penciclovir	DB00299	HSV infections	1996 (Long et al., 2012)
 Penciclovir			
Velpatasvir	DB11613	Chronic HepatitisC	2016 (Tomba et al., 2021)
 Velpatasvir			
Tenofoviridisoproxil	DB00300	HIV infection and Hepatitis B	2001 (Fischer and Ganellin, 2010)
 Tenofoviridisoproxil			
Cidofovir	DB00369	CMV retinitis	1996 (Long et al., 2012)
 Cidofovir			
Valaciclovir	DB00577	Hepatitis, HIV, and cytomegaly virus infection	2022 (Rajalakshmi et al., 2010)
 Valaciclovir			
Efavirenz	DB00625	HIV – 1infection	2016 (Vrouenraets et al., 2007)
 Efavirenz			

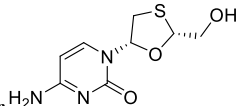
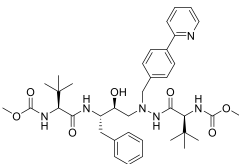
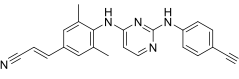
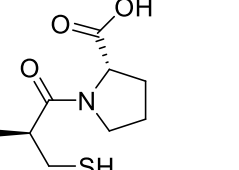
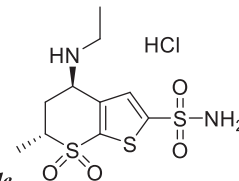
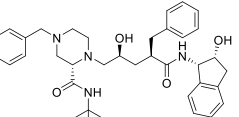
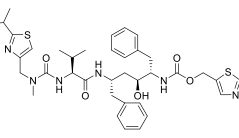
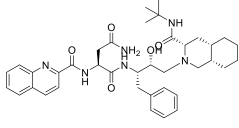
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Table 2 (continued)

Drugs	Drug Bank ID	Therapy	Approved in Year
<i>Peramivir</i> 	DB06614	InfluenzaA/B.	2014 (McLaughlin et al., 2015)
<i>Brivudine</i> 	DB03312	Herpes zoster.	2001 (Tompa et al., 2021)
<i>Maraviroc</i> 	DB04835	HIV infection	2007 (Emmelkamp and Rockstroh, 2008)
<i>Stavudine</i> 	DB00649	HIVinfection	1996 (Hill et al., 2007)
<i>Paritaprevir</i> 	DB09297	Chronic Hepatitis C	2019 (Keating, 2016)
<i>Indinavir</i> 	DB00224	HIV infection	1996 (Fischer and Ganellin, 2010)

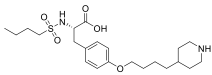
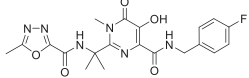
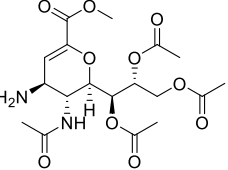
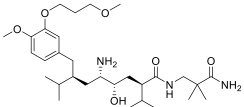
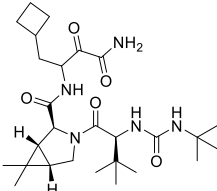
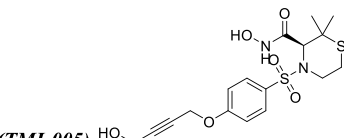
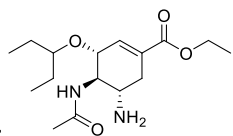
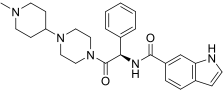
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Table 2 (continued)

Drugs	Drug Bank ID	Therapy	Approved in Year
Lamivudine	DB00709	HIV–1and hepatitis B virus (HBV)infection	1995 (Perry and Faulds, 1997)
			
Atazanavir	DB01072	HIV infection	2003 (Orrick and Steinhart, 2004)
			
Rilpivirine	DB08864	HIV–1infection	2021 (Taki et al., 2022)
			
Captopril	DB01197	Antihypertensive	1981 (Ferguson and Vlasses, 1981)
			
Dorzolamide	DB00869	Carbonic anhydrase inhibitor	1995 (Talele et al., 2010)
			
Indinavir	DB00224	Human immunodeficiency virus (HIV)	1996 (Vijayakrishnan, 2009)
			
Ritonavir	DB00503	Human immunodeficiency virus (HIV)	1996 (Van Drie, 2007)
			
Saquinavir	DB01232	Human immunodeficiency virus (HIV)	1995 (Hartman et al., 1992)
			
Tirofiban	DB00775	Fibrinogen antagonist	1998 (Schames et al., 2004)

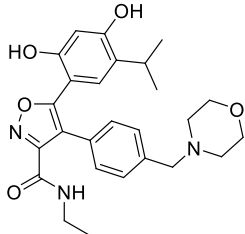
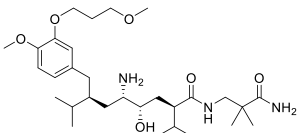
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Table 2 (continued)

Drugs	Drug Bank ID	Therapy	Approved in Year
 Tirofiban			
 Raltegravir	DB06817	Human immunodeficiency virus (HIV)	2007 (Kim et al., 1997)
 Zanamivir	DB00558	Neuraminidase inhibitor	1999 (Cohen, 2007)
 Aliskiren	DB09026	Human renin inhibitor	2007 (Kappert et al., 2008)
 Boceprevir	DB08873	Hepatitis C virus (HCV) inhibitor	2015 (Boyd et al., 2017)
 Apratastat (TMI-005)	DB13020	Rheumatoid arthritis	phase II clinical trials (Levin et al., 2006)
 Oseltamivir	DB00198	Active against influenza A and B viruses.	1999 (An et al., 2009)
 LY-517717	DB05713	Serine protease Inhibitor	2007 (Agnelli et al., 2007)phase II clinical trials

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Table 2 (continued)

Drugs	Drug Bank ID	Therapy	Approved in Year
Luminespib (NVP-AUY922)  Luminespib (NVP-AUY922)		Inhibitor for HSP90	2012 (Katerina and Evangelia, 2014) clinical trials
Aliskiren  Aliskiren	DB09026		

5.2. Virtual screening (VS)

Virtual screening identifies the most probable small molecules that bind to a drug target by computationally screening large libraries of small molecules. Structure-based virtual screening (SBVS) models use a 3-dimensional crystallographic structure of the target to dock and score compounds. Ligand-based approaches use known active compounds to search for molecules with structurally similar components (Sliwoski et al., 2014; Tan et al., 2015). Examples of successful uses in addition to the viral and antihypertensive focus of this review, include SBVS studies that identified novel scaffolds. Structures were optimized to potent inhibitors for cancer therapy by targeting the anaplastic lymphoma kinase (ALK) crystal structure (Wang and Guan, 2021). Additionally, the phytochemical libraries were virtually screened against SARS-CoV-2 targets, including the main protease (Mpro/3CLpro) and the RNA-dependent RNA polymerase (RdRp), to provide multiple leads based on natural products with better in silico binding affinity and ADMET profiles (Elkadeed et al., 2023; Elfiky, 2021)

5.3. Synergy of approaches

The combination of repurposing and virtual screening creates a great iterative loop. A hit from a repurposing screen can be used as a pharmacophore model for ligand-based virtual screening to yield potentially more novel analogs. Reciprocally, a hit from a potent virtual screen of a chemical library may have models searched in approved drugs for a repurposing candidate. The combined use of computational approaches greatly expands the application space and improves the efficiency of drug discovery, building a healthy pipeline for addressing both chronic disease and emergent infectious threats.

6. Conclusion

This extensive review highlights the impact of structure-based drug design (SBDD) and computational modeling in the development of contemporary, novel antihypertensive and antiviral agents. In-depth analyses of particular drugs, including captopril, dorzolamide, aliskiren, as well as the antiviral drugs saquinavir, ritonavir, indinavir, amprenavir, oseltamivir, and nirmatrelvir/ritonavir (Paxlovid), provide an example of rational drug design that has relied on molecular docking, QSAR, SAR, and ADMET profiling to optimize drug discovery. The synthetic pathways presented here highlight the need for stereoselectivity and optimizing functional groups to maintain binding affinity,

pharmacokinetics, and safety. The combined approach of computational methods with medicinal chemistry has facilitated the explicit targeting of important enzymes such as carbonic anhydrase, renin, ACE, HIV-1 protease, neuraminidase, and the SARS-CoV-2 main protease. The real potency of these computational methods has led to clinical success: both captopril and aliskiren became the first line agents for the treatment of hypertension, dorzolamide was an effective topical therapy for treating glaucoma while minimizing systemic side effects, and the development of many HIV protease and neuraminidase inhibitors (like oseltamivir) significantly changed treatment for patients with AIDS and influenza, respectively. Most recently, Paxlovid was introduced on the basis of existing studies targeting HIV protease and viral proteases in the course of a pandemic signifies the effectiveness of this mechanism. The review also emphasizes that drug repurposing and virtual screening have become crucial aspects of modern drug discovery. The repurposing and the virtual screening approaches utilize available chemical and clinical data to facilitate rapid identification of new therapeutic indications, with even shorter timelines and cost savings to bring forward a new indication. While repurposing drugs during the COVID-19 pandemic did not always yield desired clinical outcomes, the data collected during these attempts improved the value of the in silico computations, as they address the challenges of iterative validation of in silico predictions with clinical outcomes. We will continue to see integrated synthetic, structural, computational, and bioinformatic strategies at heart of designing effective therapeutics for the continually shifting panorama of new infectious diseases and increasingly complex chronic conditions. While the distance from a computational model to a clinically approved drug remains daunting, it is abundantly clear from the review that in modern medicinal chemistry, synergy of predictive in silico tools and equally diligent experimental validation is the foundation.

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CRediT authorship contribution statement

Asmaa M. Fahim: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Not Applicable.

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